

## RESEARCH ARTICLE

A Census of Double-Peaked Ly $\alpha$  Emitters in MAGPI: Classification, Global Characteristics, and Spatially Resolved Properties

Tamal Mukherjee<sup>1,2,3</sup> Tayyaba Zafar<sup>1,2,3</sup> Themiya Nanayakkara<sup>4,2</sup> Siddhartha Gurung-Lopez<sup>5,6</sup> Anshu Gupta<sup>7,2</sup> Scott M. Croom<sup>8,2</sup> Andrew Battisti<sup>9,10,2</sup> Karl Glazebrook<sup>4,11</sup> Polychronis Papaderos<sup>12,13</sup> Melissa Riggs<sup>1,3</sup> Emily Wisnioski<sup>10,2</sup> Caroline Foster<sup>14,2</sup> Katherine E. Harborne<sup>15,16,17</sup> Claudia D. P. Lagos<sup>9,2</sup> J. Trevor Mendel<sup>10,2</sup> Jahang Prathap<sup>1,2,3</sup> Stefania Barsanti<sup>18,10,2</sup> Sarah M. Sweet<sup>19,2</sup> Lucas M. Valenzuela<sup>20</sup> and Anilkumar Mailvaganam<sup>1,2,3</sup>

<sup>1</sup>School of Mathematical and Physical Sciences, Macquarie University, NSW 2109, Australia

<sup>2</sup>ARC Centre of Excellence for All Sky Astrophysics in 3 Dimensions (ASTRO 3D)

<sup>3</sup>Astrophysics and Space Technologies Research Centre, Macquarie University, Sydney, NSW 2109, Australia

<sup>4</sup>Centre for Astrophysics and Supercomputing, Swinburne University of Technology, PO Box 218, Hawthorn 3122, VIC, Australia

<sup>5</sup>Observatori Astronòmic de la Universitat de València, Ed. Instituts d'Investigació, Parc Científic. C/ Catedrático José Beltrán, n2, 46980 Paterna, Valencia, Spain

<sup>6</sup>Departament d'Astronomia i Astrofísica, Universitat de València, 46100 Burjassot, Spain

<sup>7</sup>International Centre for Radio Astronomy Research (ICRAR), Curtin University, Bentley, WA, Australia

<sup>8</sup>Sydney Institute for Astronomy, School of Physics, University of Sydney, NSW 2006, Australia

<sup>9</sup>International Centre for Radio Astronomy Research (ICRAR), The University of Western Australia, Crawley, WA 6009, Australia

<sup>10</sup>Research School of Astronomy and Astrophysics, Australian National University, Canberra, ACT 2611, Australia

<sup>11</sup>JWST Australian Data Centre (JADC), Swinburne Advanced Manufacturing and Design Centre (AMDC), John Street, Hawthorn, VIC 3122, Australia

<sup>12</sup>Centro de Astrofísica e Ciências do Espaço, Universidade de Lisboa - OAL, Tapada da Ajuda, PT1349-018 Lisboa, Portugal

<sup>13</sup>Instituto de Astrofísica e Ciências do Espaço - Centro de Astrofísica da Universidade do Porto, Rua das Estrelas, 4150-762 Porto, Portugal

<sup>14</sup>School of Physics, University of New South Wales, Sydney, NSW 2052, Australia

<sup>15</sup>Institute for Computational Cosmology, Durham University, South Road, Durham DH1 3LE, UK

<sup>16</sup>Centre for Extragalactic Astronomy, Durham University, South Road, Durham DH1 3LE, UK

<sup>17</sup>Department of Physics, Durham University, South Road, Durham DH1 3LE, UK

<sup>18</sup>Sydney Institute for Astronomy, School of Physics, University of Sydney, NSW 2006, Australia

<sup>19</sup>School of Mathematics and Physics, University of Queensland, Brisbane, QLD 4072, Australia

<sup>20</sup>Universitäts-Sternwarte, Fakultät für Physik, Ludwig-Maximilians-Universität München, Scheinerstr. 1, 81679 München, Germany

Author for correspondence: T. Mukherjee, Email: tamal.mukherjee@hdr.mq.edu.au.

## Abstract

Double-peaked Lyman- $\alpha$  (Ly $\alpha$ ) profiles provide critical insights into gas kinematics and the distribution of neutral hydrogen (H I) from the interstellar to the intergalactic medium (ISM to IGM), and serve as valuable diagnostics of ionising Lyman continuum (LyC) photon escape. We present a study of the global and spatially resolved properties of double-peaked Ly $\alpha$  emitters (LAEs), based on VLT/MUSE data from the Middle Ages Galaxy Properties with Integral Field Spectroscopy (MAGPI) survey. From a parent sample of 417 LAEs at  $z = 2.9 - 6.6$  in the first 35 fields, we identify 108 double-peaked LAEs using an automated peak classification technique. We measure a double-peak fraction of  $\sim 37\%$  at  $z < 4$ , decreasing to  $\sim 14\%$  at  $z > 4$ , likely due to enhanced IGM attenuation. Approximately 17% of the double-peaked LAEs are blue-dominated, suggesting gas inflows. The blue-to-total flux ratio exhibits a luminosity dependence: fainter lines generally show higher blue flux, though a few luminous sources also show strong blue peaks. We find a significant narrowing of the red peak at  $z > 4$ , despite the presence of the blue peak, indicating intrinsic galaxy evolution rather than an effect of IGM attenuation. Several LAEs exhibit residual flux in the absorption trough, with normalised trough flux anticorrelating with peak separation, reflecting variations in H I column density. We further investigate spatially-resolved properties of ten red-dominated LAEs with extended Ly $\alpha$  halos. Despite azimuthal variations, both the blue-to-total flux ratio and normalised trough flux density increase with radius, while peak separation decreases. The red peak asymmetry shows only minor radial changes. These trends are consistent with variations in shell outflow velocity and H I column density across the halos. However, some exceptions to these patterns are also noted. Based on peak separation, red peak asymmetry, and residual trough flux, we identify five LAEs as strong LyC leaker candidates.

**Keywords:** galaxies: high redshift – galaxies: formation – galaxies: evolution – cosmology: observations

## 1. Introduction

The Ly $\alpha$  emission (1215.67 Å) line of hydrogen atom is a key spectral feature that is used to detect and study high-redshift galaxies (e.g., [Dijkstra, 2014](#); [Hayes, 2015](#); [Ouchi et al., 2020](#)). Due to their resonant nature, Ly $\alpha$  photons scatter frequently in the interstellar, circumgalactic, and even intergalactic medium (ISM, CGM, and IGM, respectively). This random walk leads

to a wide diversity of spectral shapes of Ly $\alpha$  that contain information about the kinematics and geometry of the gas present in ISM and CGM (see, [Dijkstra et al. 2019](#) for a review). The characteristic shape of Ly $\alpha$  emission, making it easily recognisable, is typically a single red asymmetric line profile (e.g., [Shapley et al., 2003](#); [Tapken et al., 2007](#)). However, over the past decade, double-peaked profiles of Ly $\alpha$ , characterised by

two asymmetric peaks, have been detected in both local galaxies (e.g., Jaskot & Oey, 2014; Henry et al., 2015; Verhamme et al., 2017; Orlitová et al., 2018; Izotov et al., 2018) and galaxies out to high redshifts (i.e.,  $z = 2 - 7$ ; Kulas et al., 2012; Yamada et al., 2012; Matthee et al., 2018; Hayes et al., 2021; Meyer et al., 2021; Songaila et al., 2022; Erb et al., 2023; Moya-Sierralta et al., 2024; Vitte et al., 2025). Double-peaked profiles are also detected in Lyman alpha nebulae, or blobs (Yang et al., 2014; Vanzella et al., 2017; Li et al., 2022), suggesting complex radiative transfer effects and the presence of large-scale gas kinematics. In addition, a few triple-peaked profiles have also been reported (e.g., Rivera-Thorsen et al., 2017; Vanzella et al., 2018; Vitte et al., 2025). The Multi-Unit Spectroscopic Explorer (MUSE) at the Very Large Telescope (VLT; Bacon et al., 2010), has revealed a vast population of Ly $\alpha$  emitters (LAEs) with diverse spectral profiles at redshifts  $z = 2.9 - 6.7$  (see, Herenz et al., 2017; Bacon et al., 2023; Vitte et al., 2025). Meanwhile, the Near Infrared Spectrograph aboard the James Webb Space Telescope (JWST/NIRSpec) is already extending the boundaries of Ly $\alpha$  line observations to even higher redshifts (Bunker et al., 2024).

Double-peaked Ly $\alpha$  profiles typically arise from two scenarios: (1) scattering of photons from a monochromatic or continuum source in a static, dust-free neutral hydrogen cloud, producing roughly two symmetric peaks; or (2) scattering of photons from a central source in an expanding or infalling medium. In the latter case, in an outflowing medium, photons preferentially escape on the red side, producing a dominant red peak with a weaker blue bump. Additionally, Ly $\alpha$  photons can backscatter off a receding outflow on the far side of the galaxy, which shifts their frequency out of resonance and allows them to escape without further scatterings (Dijkstra et al., 2006; Verhamme et al., 2006; Steidel et al., 2010), while multiple backscattering events can generate additional redshifted peaks and broad, extended red wings (Verhamme et al., 2006). A blueshifted peak usually appears when either the outflow has low opacity (e.g., lower column density or higher ionisation), enabling some photons to escape directly or when photons scatter off of the neutral hydrogen (H I) gas in an inflowing CGM (Zheng & Miralda-Escudé, 2002; Dijkstra et al., 2006). Moreover, a dominant blueshifted peak over a redshifted peak arises when accretion/inflow dominates over outflow (see Mukherjee et al. 2023; Bolda et al. 2024, and references therein). Alternatively, double-peaked Ly $\alpha$  profiles can also arise from two distinct, closely separated galaxies, each contributing its own Ly $\alpha$  emission line peak with a velocity offset (Cooke et al., 2010). In the local universe, the separation between the two Ly $\alpha$  peaks is found to correlate with the escape of ionising Lyman-continuum (LyC) radiation, potentially due to low column density channel of H I (e.g., Verhamme et al., 2017; Izotov et al., 2018; Flury et al., 2022). Direct observations of the escaping LyC photons become nearly impossible at  $z > 4$  due to significant attenuation by neutral hydrogen in the IGM (see Ouchi et al. 2020; and references therein). Therefore, the double-peaked Ly $\alpha$  profile emerges as an indirect tracer of LyC escape, as well as the outflow and inflow kinematics that regulate the escape of both

Ly $\alpha$  and LyC photons.

The theory describing Ly $\alpha$  radiative transfer has been studied for decades and for a range of gas geometries. A variety of powerful public codes have enabled numerical solutions to the Ly $\alpha$  radiative transfer problem. The most widely used radiative transfer model is typically modeling outflows as spherical, expanding shells around point Ly $\alpha$  emitting region (e.g., Verhamme et al., 2008, 2015; Gronke et al., 2015; Gurung-López et al., 2022; Gurung-Lopez et al., 2025). Despite their simple geometry, these models successfully reproduce various observed Ly $\alpha$  profiles, including single and double-peaked features seen in Lyman break galaxies, LAEs, damped Ly $\alpha$  systems, and Green Pea galaxies (e.g., Verhamme et al., 2008; Vanzella et al., 2010; Hashimoto et al., 2015; Gronke, 2017; Yang et al., 2017) and constrained properties of the scattering medium, such as H I column density and shell velocity (e.g., Verhamme et al., 2015; Gronke et al., 2017). In general, a lower shell velocity enhances the blue peak, while a lower H I column density reduces the peak separation (Verhamme et al., 2015). However, physical parameters derived from the shell model often differ from those constrained by interstellar absorption and nebular emission lines (Kulas et al., 2012; Orlitová et al., 2018). Recent studies have expanded Ly $\alpha$  radiative transfer to multiphase, clumpy media, where cool H I clumps are embedded in a hot, ionised inter-clump medium (e.g., Neufeld, 1991; Hansen & Oh, 2006; Laursen et al., 2013; Gronke & Dijkstra, 2016; Li & Gronke, 2022).

Ly $\alpha$  halos or extended Ly $\alpha$  emission, extending several kiloparsecs around galaxies, trace CGM gas distribution, and kinematics through resonant scattering. Extended Ly $\alpha$  halos are generally interpreted as arising from resonant scattering of Ly $\alpha$  photons produced in central star-forming regions, rather than in-situ recombination at large radii. This is evident from local-universe studies (e.g., Lyman Alpha Reference Sample or LARS) showing Ly $\alpha$  emission extending 2 to 4 times farther than H $\alpha$  (Hayes et al., 2013; Östlin et al., 2014), and by high- $z$  statistical analyses indicating that halo properties correlate with central Ly $\alpha$  luminosity rather than halo mass (Momose et al., 2014; Kusakabe et al., 2019). However, detecting Ly $\alpha$  halos at  $z \gtrsim 2$  is challenging due to their faint emission and sensitivity limits. In recent years, IFU surveys such as MUSE and the Keck Cosmic Web Imager (KCWI; Martin et al., 2010) have enabled the detailed study of individual halos around galaxies from  $2 \lesssim z \lesssim 6$  (see, Wisotzki et al., 2016; Leclercq et al., 2017; Erb et al., 2018; Leclercq et al., 2020; Erb et al., 2023; Weldon et al., 2024). So far, MUSE studies have analysed the spectral properties across the halo around single-peaked LAEs, finding a correlation between the line width and velocity shift of the peak, with broader emission often tending to come from the outer regions of halo (Claeyssens et al., 2019; Leclercq et al., 2020). On the other hand, Erb et al. (2023) and Weldon et al. (2024) study variations in the peak ratio and separation of the double-peaked Ly $\alpha$  profile across the extended halos detected in KCWI, and find that peak separation decreases and the blue-to-red flux ratio increases toward the outskirts of the halo.

In this paper, we present a large sample of double-peaked

Ly $\alpha$  emitters identified through MUSE observations from the Middle Ages Galaxy Properties with Integral Field Spectroscopy (MAGPI) survey (Foster et al., 2021), and provide a comprehensive analysis of their spectroscopic properties. In addition to characterising the global spectral features, we conduct a spatially resolved analysis of the double-peaked Ly $\alpha$  profiles across ten extended halos at  $z > 3$ . To interpret these observations, we employ radiative transfer modeling to study the underlying gas kinematics and physical conditions driving Ly $\alpha$  line formation.

The paper is structured as follows: In §2, we briefly describe the MAGPI observations, data reduction and the method of identifying LAEs from datacubes. §3 outlines our automated classification method for diverse Ly $\alpha$  spectral shapes and the selection of double-peaked LAEs. In §4, we present a forward-modeling approach to fit the observed double-peaked profiles while accounting for the instrumental resolution. In §5, we present both global and spatially resolved spectroscopic properties of our sample. §6 discusses these results in the context of radiative transfer models. We summarise our conclusions in §7. Throughout, we assume a flat  $\Lambda$ CDM cosmology with  $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_m = 0.3$ , and  $\Omega_\Lambda = 0.7$ .

## 2. Data

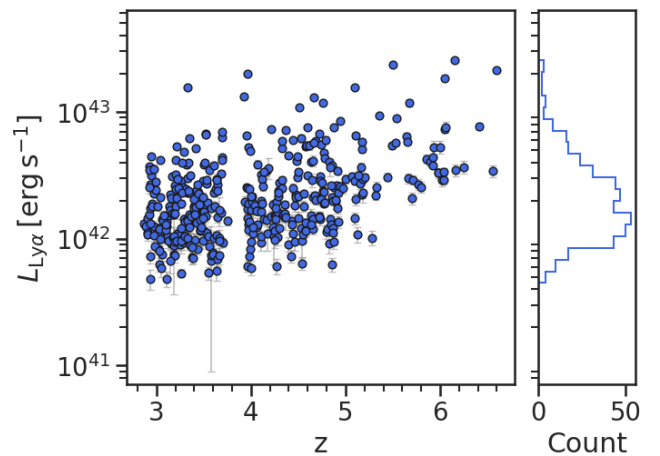
### 2.1 Observations and data reduction

The MAGPI survey<sup>1</sup> is an ongoing Large Program on the VLT/MUSE, targeting 56 fields from the Galaxy and Mass Assembly (GAMA; Driver et al., 2011) G12, G15 and G23 fields. MAGPI also includes archival observations of legacy fields Abell 370 and Abell 2744. The survey targets a total of 60 primary galaxies with stellar masses  $M_* > 7 \times 10^{10} M_\odot$  and  $\sim 100$  satellite galaxies with  $M_* > 10^9 M_\odot$ . The primary objective of MAGPI is to conduct a detailed spatially resolved spectroscopic analysis of stars and ionised gas within  $0.25 < z < 0.35$  galaxies (see Foster et al., 2021). Data are taken using the MUSE Wide Field Mode ( $1' \times 1'$ ) with a spatial sampling rate of  $0.2''/\text{pixel}$  and the median Full Width at Half Maximum (FWHM) is  $0.64''$ ,  $0.6''$  and  $0.55''$  in  $g$ ,  $r$  and  $i$  band respectively. Each field is observed in six observing blocks, each comprising  $2 \times 1320$  s exposures, resulting in a total integration time of 4.4 h. The survey primarily employs the nominal mode (data are taken with the blue cut-off filter in place), providing a wavelength coverage ranging from  $4700 \text{ \AA}$  to  $9350 \text{ \AA}$ , with a dispersion of  $1.25 \text{ \AA}$ . Ground-layer adaptive optics (GLAO) is used to correct atmospheric seeing effects, resulting in a gap between  $5805 \text{ \AA}$  and  $5965 \text{ \AA}$  due to the GALACSI laser notch filter. The MUSE data reduction process is thoroughly described in Foster et al. (2021).

### 2.2 LAE identification

The depth of the MAGPI data enables the detection of both foreground sources within the Local Universe and distant

background sources, including LAEs in the redshift range  $2.9 \lesssim z \lesssim 6.6$ . To identify emission lines, we employ LSDCat 1.0<sup>2</sup> (Herenz & Wisotzki, 2017), a Python-based tool designed to detect faint emission lines in integral-field spectroscopic datacubes by correlating a matched filter with the data. A detection threshold of  $\text{SN}_{\text{thresh}} = 7.0$  was adopted to minimize false positives. This choice is in line with thresholds used in other MUSE-based LAE searches; for instance, the MUSE-WIDE survey applies a threshold of  $5 - 6.4$  for LAE detection (see Kerutt et al., 2022). The initial list of emission-line candidates is then passed through the LAE\_Scanner<sup>3</sup> tool, which filters out spurious and low-significance detections by enforcing criteria on spectral shape, substantially reducing the false-positive rate. For the remaining candidates, segmentation maps were created using the PROFOUND R package (Robotham et al., 2018) and then global spectra are extracted. Redshifts are estimated using the MARZ<sup>4</sup> software (Hinton et al., 2016), which provides automated redshift identification based on spectral templates. During this step, all candidates undergo visual inspection to confirm the presence and shape of the Ly $\alpha$  line, and to exclude contaminants such as low-redshift [O II] emitters by checking whether the observed line profile matches the expected [O II]  $\lambda 3726, 3729$  doublet separation at lower redshifts. Following these steps, we identified 417 LAEs in the first 35 MAGPI fields. A circular aperture of  $1.4''$  radius is used to compute spectroscopic properties, chosen to match the typical extent of LAEs while minimizing contamination from noisy spaxels. A redshift distribution of Ly $\alpha$  luminosities is presented in Figure 1.



**Figure 1.** Ly $\alpha$  luminosity as a function of spectroscopic redshift for 417 LAEs identified in first 35 MAGPI fields, spanning the redshift range  $2.9 \lesssim z \lesssim 6.6$ .

## 3. Peak classification and sample selection

In this section, we outline the process of selecting double-peaked profiles from the MAGPI LAEs. Given that the observed Ly $\alpha$  line profiles can show multiple peaks, we develop

<sup>1</sup>Based on observations obtained using MUSE instrument at VLT of the European Southern Observatory (ESO), Paranal, Chile (ESO program ID 1104.B – 0536)

<sup>2</sup><https://bitbucket.org/Knuser2000/lscat/>

<sup>3</sup>[https://github.com/robbassett/LAE\\_scanner](https://github.com/robbassett/LAE_scanner)

<sup>4</sup><https://github.com/Samreay/Marz>

an automated method to systematically identify and characterise spectral features throughout the entire MAGPI LAE sample. The process involves the following steps:

- We first convert the observed wavelengths to vacuum wavelengths using the `pyasl` module from the `PyAstronomy` package. We then use a python package `pyplatefit`<sup>5</sup> (Bacon et al., 2023) to fit and subtract any stellar continuum from each spectrum. Finally, we convert these spectra from wavelength to velocity space, over a velocity window of  $\pm 2000$  km/s, ensuring that this covers the entire Ly $\alpha$  emission (Kerutt et al., 2022; Vitte et al., 2025).
- We take the continuum-subtracted Ly $\alpha$  spectra and obtain S/N spectra using the corresponding  $1\sigma$  uncertainties on flux densities. We smooth the S/N spectra with a 1-pixel moving average to reduce the impact of isolated noisy pixels. We then run `find_peaks` from `SciPy.signal` on the S/N spectra with a threshold of 1.5 to detect faint local maxima. For each local maxima, we locate the nearest local minima on either side and confirm whether at least two adjacent pixels in that window have  $S/N \geq 1.5$ . If this holds, we mark those regions as significant with the maxima as possible peaks.
- We then fit a multi-asymmetric Gaussian function to the original spectra, focusing only on the regions around potential peaks and fixing the Gaussian mean to the corresponding peak velocity. The fitting function is defined as:

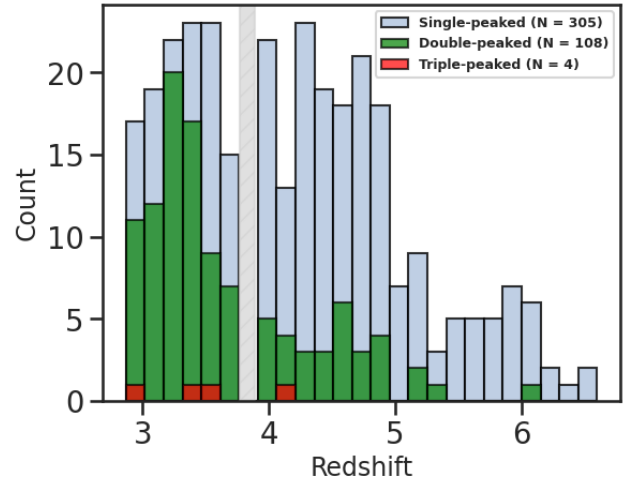
$$F(\nu) = \sum_{i=1}^N A_i \exp\left(-\frac{\Delta\nu_i^2}{2(a_{\text{asym},i} \Delta\nu_i + w_i)^2}\right) \quad (1)$$

where  $N$  is the number of potential peaks from the previous step,  $A_i$  is the peak amplitude,  $\Delta\nu_i = \nu - \nu_{0,i}$  is the velocity offset (in  $\text{km s}^{-1}$ ),  $a_{\text{asym},i}$  controls the asymmetry (or skewness), and  $w_i$  is the line width (in  $\text{km s}^{-1}$ ). We perform the fit using `curve_fit` from the `scipy.optimize` package, incorporating the  $1\sigma$  flux uncertainties. During fitting, we impose two criteria: (i) the integrated S/N of each Gaussian must be  $\geq 1.5$ ; (ii) the FWHM must exceed the wavelength-dependent line spread function (LSF). We compute LSF for each MAGPI field using a second-order polynomial of the form:  $\text{FWHM}(\lambda) = A\lambda^{-2} + B\lambda^{-1} + C$  (see e.g., Kerutt et al., 2022). This step results in the detection of up to  $N = 3$  significant peaks, classified as single-, double-, or triple-peaked profiles for  $N = 1, 2$ , and 3, respectively.

- We then spatially verify those candidates using Ly $\alpha$  narrow-band (NB) images. For this, we use the MUSE Python Data Analysis Framework (MPDAF<sup>6</sup>) package to generate NB images for each peak and compare centroids. In this process, we apply a signal-to-noise threshold of  $S/N = 1.5$  to the NB images. A secondary peak is considered co-spatial if its NB centroid falls within a  $0.6''$  radius aperture

centered on the centroid of the primary peak (i.e., the peak with the highest integrated flux). This step reclassifies peaks based on their spatial origin, as the two peaks in some double-peaked profiles arise from distinct spatial locations, indicating that the observed structure is not the result of radiative transfer effects. Therefore, we retain only the co-spatial peaks, resulting in a final classification of 108 double-peaked, 4 triple-peaked, and the remaining 306 classified as single-peaked profiles (Figure 2).

A comprehensive analysis of all MAGPI LAE profiles will be presented in Mukherjee et al. (in prep). Here, we focus exclusively on the 108 confirmed double-peaked LAEs, representing  $\sim 26\%$  of the MAGPI LAEs. However, this fraction can be influenced by selection biases, including survey depth and the adopted S/N threshold. At  $z < 4$ , we find a double-peaked fraction of  $\sim 37\%$ , consistent with previous studies: 30% by Kulas et al. (2012), 25% by Sobral et al. (2018), 32.9% for MUSE-Deep, and 33.8% for MUSE-Wide by Kerutt et al. (2022). As shown in Figure 2, the distribution of double-peaked LAEs peaks at  $z \sim 3.2$  and rapidly drops beyond  $z > 4$ . This decline is expected due to the increasing neutral fraction of the IGM at higher redshifts (Hayes et al., 2021). At  $z > 4$ , we find the double-peak fraction to be only  $\sim 14\%$ , consistent with the fraction reported for other MUSE observations (Kerutt et al., 2022) at similar redshifts.



**Figure 2.** Redshift distribution of MAGPI LAEs classified by their Ly $\alpha$  profile as outlined in §3: single-peaked (light-blue), double-peaked (green), and triple-peaked (red). The gray-shaded region ( $z = 3.75\text{--}3.9$ ) marks a spectral gap due to the GALACSI laser notch filter, resulting in a lack of detected sources in this range.

#### 4. Forward modeling

In this section, we introduce a forward modeling approach based on double-asymmetric Gaussian (DAG) functions to characterise the double-peaked profiles observed in our sample. The idea of a forward model is to take into account the MUSE instrumental broadening. To accurately infer the intrinsic properties of the Ly $\alpha$  line profiles, it is crucial to account

<sup>5</sup><https://github.com/musevlt/pyplatefit>

<sup>6</sup><https://pypi.org/project/mpdaf/>



for this instrumental effect. Forward modeling provides a robust framework to do this. Before that, we first determine their systemic redshifts. For this, we apply the `pyplatefit` to search for C III]  $\lambda 1907$  Å emission, a known proxy for systemic redshift in LAEs (Verhamme et al., 2018). We detect C III] emission with a  $> 2\sigma$  significance in 14 sources at  $z = 2.9$ – $3.7$ , and adopt their C III] redshifts as systemic redshift ( $z_{\text{sys}}$ ). An example of this is presented in Figure 3. Unfortunately, with the current MUSE data we are limited in the precision we can achieve for these faint emission lines. Therefore, we caution that the reported systemic redshifts should be considered as indicative rather than definitive. For LAEs without C III] detection, we use redshift of the absorption trough between the two Ly $\alpha$  peaks as a proxy for  $z_{\text{sys}}$ . In the forward modeling approach:

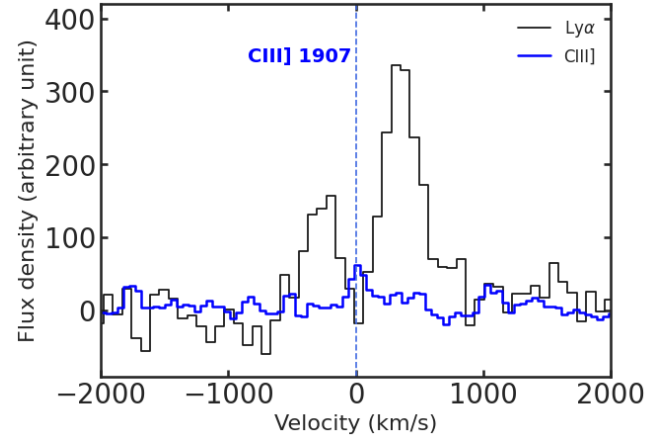
- We begin with a double-asymmetric Gaussian (i.e.  $N = 2$  in Eq. 1), with eight parameters in total. These parameters represent the intrinsic, pre-convolution state of the emission line, which is what the profile would look like without instrumental effects.
- This model is then convolved with the wavelength dependent LSF, assumed to be Gaussian (with FWHM as mentioned in §3) to match the spectral resolution of the data. This convolution accounts for the spectral resolution of MUSE and models the blurring introduced by the instrument.
- Finally, we fit this convolved model to the observed double-peaked profiles. For this, we first used the `curve_fit` routine to perform a non-linear least squares fit to the observed spectrum. The purpose of this initial fit is to obtain reasonable estimates of the parameters.
- The parameter estimates from `curve_fit` is then used as the initial positions for the walkers in the MCMC sampling, performed using the `emcee`<sup>7</sup> package. MCMC explores the parameter space to derive the probability distribution. By setting `is_weighted = True`, we force the `emcee` to directly sample the posterior distributions of parameters with the correct weighting from  $1\sigma$  uncertainty array, obtained from the variance cube. We use a sufficiently large chain length (steps = 10000) to ensure that the MCMC solution has converged.

Figure 4 shows how instrumental broadening affects the observed spectra. We present all the double-peaked profiles and their corresponding DAG fits in Figure 16 in the Appendix. The observed profiles are well described by these fits.

## 5. Results

We now present the analysis of the spectroscopic properties of the 108 confirmed double-peaked LAEs in our sample. In this section, we describe how key spectral measurements are obtained and outline the global and spatially resolved properties of these sources.

<sup>7</sup><https://github.com/dfm/emcee>



**Figure 3.** One example of rest-frame spectrum of a bright LAE (ID: 1201250296) with detected C III]  $\lambda 1907$  emission ( $\sim 2.75\sigma$ ) probing the systemic redshift  $z_{\text{sys}} = 3.2377$ .

### 5.1 Spectral Measurements

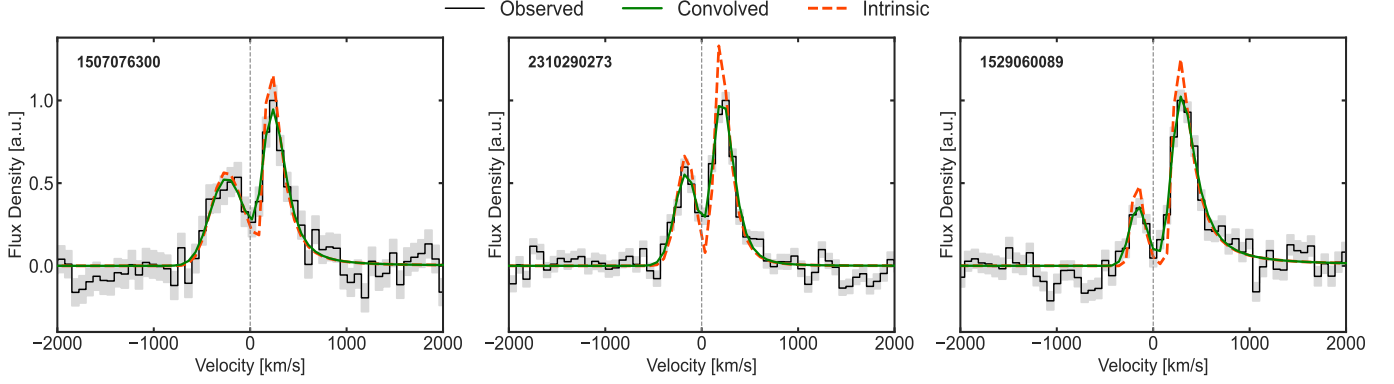
We measure the following spectroscopic quantities from the intrinsic model parameters, reflecting true profiles before instrumental broadening:

- Peak velocity separation ( $\Delta_{\text{peak}}$ ): Defined as the velocity difference between the red and blue peaks,  $\Delta_{\text{peak}} = V_{\text{red}} - V_{\text{blue}}$ , where both velocities are measured with respect to the systemic redshift, in  $\text{km s}^{-1}$ .
- Total Ly $\alpha$  flux and luminosity: The total Ly $\alpha$  flux is obtained by integrating the flux density under the area of the DAG. Flux uncertainties are drawn from the posterior distributions of the MCMC chains, providing parameter uncertainties that better reflect the full likelihood landscape. The Ly $\alpha$  luminosity,  $L_{\text{Ly}\alpha}$ , is derived from the total flux, and its corresponding errors.
- Integrated signal-to-noise ratio,  $(S/N)_{\text{total}}$ : Calculated as the total integrated Ly $\alpha$  flux divided by the flux uncertainty.
- Blue-to-total flux ratio ( $F_{\text{blue}}/F_{\text{total}}$ ): The relative strength of the blue peak is quantified as  $F_{\text{blue}}/F_{\text{total}} = F_{\text{blue}}/(F_{\text{blue}} + F_{\text{red}})$ , where  $F_{\text{blue}}$  and  $F_{\text{red}}$  are the integrated fluxes of the blue and red peaks, respectively.
- FWHM of each peak: The line width of each peak is calculated using the fitting parameters (see, Mukherjee et al., 2024):

$$\text{FWHM (km s}^{-1}\text{)} = \frac{2\sqrt{2 \ln(2)} w}{1 - 2 \ln(2) a_{\text{asym}}^2} \quad (2)$$

where  $w$  and  $a_{\text{asym}}$  are the width and asymmetry parameters of the fit, respectively. Uncertainties are derived from the errors in the fit parameter returned by `emcee`.

- Flux density in the absorption trough ( $F_{\text{trough}}$ ): measured directly from the intrinsic DAG profile as the flux density at the minimum between the two peaks, with uncertainties estimated from MCMC sampling.

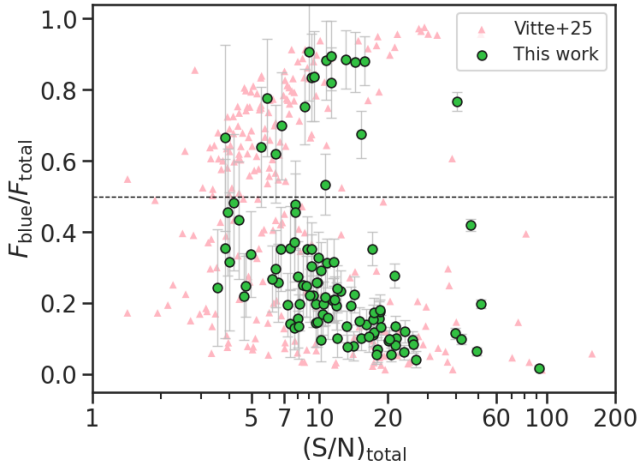


**Figure 4.** Forward modeling of three representative double-peaked Ly $\alpha$  spectra. The observed spectra are shown in black, with the best-fit forward-modeled profiles overplotted in green after convolution with the instrumental LSF. The intrinsic double asymmetric Gaussian models prior to convolution are shown in red. The corresponding MAGPI IDs are shown in the top left corner of each panel.

- Ly $\alpha$  red peak asymmetry parameter,  $A_f$ : This is defined as follows:

$$A_f = \frac{\int_{\lambda_{\text{red}}}^{\infty} f_{\lambda} d\lambda}{\int_{\lambda_{\text{trough}}}^{\lambda_{\text{red}}} f_{\lambda} d\lambda} \quad (3)$$

where  $\lambda_{\text{red}}$  and  $\lambda_{\text{trough}}$  are the wavelengths of the red peak and absorption trough, respectively (Rhoads et al., 2003; Kakiichi & Gronke, 2021).



**Figure 5.** Blue-to-total flux ratio as a function of the total S/N of the Ly $\alpha$  line in our parent sample. Pink triangles are taken from Vitte et al. (2025). The horizontal black dashed line marks the division between blue- and red-dominated regions.

## 5.2 Global Ly $\alpha$ profiles

This section describes the global properties of Ly $\alpha$  emission measured from spatially integrated spectra. Full spectroscopic measurements are listed in Table 2 in the Appendix.

### 5.2.1 Flux ratio distribution versus $(S/N)_{\text{total}}$

In our parent double-peaked LAE sample, the total integrated S/N of the Ly $\alpha$  line ranges from 3.52 to 92.11. We find

18/108 ( $\sim 17\%$ ) double-peaked profiles are blue-dominated ( $F_{\text{blue}}/F_{\text{total}} > 0.5$ ). Three of these have systemic redshifts that confirm inflows or accretion-dominated sources. Of which two bright cases showing extended halos are reported in Mukherjee et al. (2023). The remaining 90 sources are red-dominated ( $F_{\text{blue}}/F_{\text{total}} < 0.5$ ), indicating gas outflows. This blue-to-red fraction is in agreement with cosmological zoom-in simulation (Blaizot et al., 2023) of a galaxy, where  $< 20\%$  of the profiles are found to be blue-dominated. However, a recent study of LAEs from the MUSE Extremely Deep Field found a significantly higher fraction ( $> 40\%$ ) (Vitte et al., 2025). Figure 5 shows the blue-to-total flux ratio versus  $(S/N)_{\text{total}}$ , including data from Vitte et al. (2025). We find blue peak flux increases with decreasing  $(S/N)_{\text{total}}$ , while extreme blue-dominated LAEs with  $F_{\text{blue}}/F_{\text{total}} > 0.8$  are only detected for  $(S/N)_{\text{total}} \gtrsim 7$ .

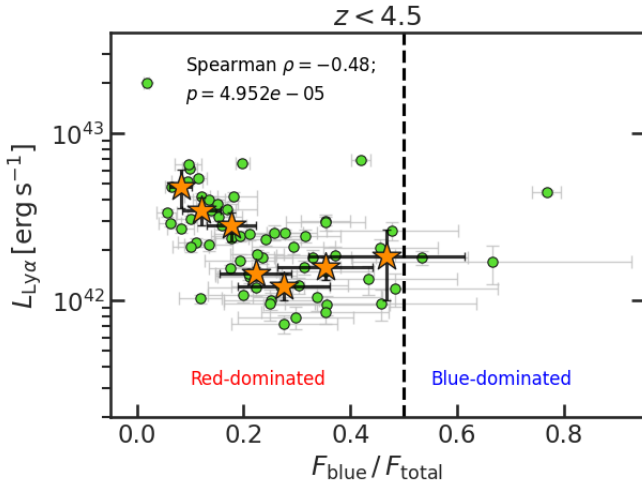
**Table 1.** Summary of the double-peaked LAE sample used in this work.

| Sample              | Selection criterion                     | Number of sources |
|---------------------|-----------------------------------------|-------------------|
| Parent sample       | All detected double peaks               | 108               |
| Conservative sample | Both peaks with integrated S/N $\geq 3$ | 76                |

For robust measurements, we adopt a conservative threshold of  $(S/N)_{\text{blue}} \geq 3$  and  $(S/N)_{\text{red}} \geq 3$  throughout the rest of the paper, resulting in a “Conservative sample” of 76 sources (see Table 1).

### 5.2.2 Ly $\alpha$ luminosity versus flux ratio

In Figure 6, we study the relation between Ly $\alpha$  luminosity and  $F_{\text{blue}}/F_{\text{total}}$  for our conservative sample. For this, we take sources only up to  $z < 4.5$ . Sources at higher redshift ( $z > 4.5$ ) are excluded, as they are strongly affected by IGM absorption and tend to be more luminous, which could bias the relation. We find a strong anticorrelation (Spearman  $\rho = -0.48$ , p-value =  $4.95 \times 10^{-5}$ ) between Ly $\alpha$  luminosity and  $F_{\text{blue}}/F_{\text{total}}$  (Figure 6): Brighter Ly $\alpha$  profiles show stronger red peaks, while fainter ones have more blue peak flux. We perform a quantile-based binning, taking same number of



**Figure 6.** Observed  $\text{Ly}\alpha$  luminosity versus  $F_{\text{blue}}/F_{\text{total}}$  for 64 LAEs in our conservative sample at  $z < 4.5$ , where the IGM has less effect. The vertical black dashed line separates the blue-dominated and red-dominated regions. Orange stars represent the median luminosity within each of the 7 equal-population bins (9 sources per bin), with the scatter in each bin indicated by black error bars.

sources within each  $F_{\text{blue}}/F_{\text{total}}$  bin. We find that in the red-dominated region, the bin median luminosity decreases as  $F_{\text{blue}}/F_{\text{total}}$  increases (see Figure 6). This trend is consistent with cosmological simulations (Blaizot et al., 2023) using mock  $\text{Ly}\alpha$  spectra. Blaizot et al. (2023) identify this relation in a single simulated galaxy with a relatively constant stellar mass and star formation rate (SFR) during its evolutionary phase. Our results extend this trend to a broader population of double-peaked LAEs across a wide redshift range.

The relation flattens at lower luminosities and higher ratios, likely because, in the blue-dominated regime (Figure 6), inflows or gas accretion become the dominant mechanism driving  $\text{Ly}\alpha$  emission, contributing significantly to the luminosity.

### 5.2.3 Redshift distribution of line widths

We find mean FWHMs of  $226.03 \pm 125.15 \text{ km s}^{-1}$  and  $225.86 \pm 42.80 \text{ km s}^{-1}$  for the blue and red peak, respectively. In Figure 7, we investigate the redshift dependence of the FWHM for both the red and blue peaks of the  $\text{Ly}\alpha$  line profile. We find that at  $z < 4$ , the red peak exhibits a wide range of line widths, including both narrow and broad profiles (see the left panel of Figure 7), indicating a greater diversity in kinematic conditions or radiative transfer effects at these redshifts. In contrast, at  $z \gtrsim 4$ , the FWHM of the red peak shows a tendency to decrease with increasing redshift. We compare our data with that of Kerutt et al. (2022) and notice a comprehensive trend for MUSE LAEs. This narrowing is not seen with high significance in the FWHM of the blue peak (right panel). However, at  $z > 5.5$ , the trend is influenced by limited statistics, where the likelihood of observing a double peak becomes nearly negligible due to the impact of the IGM. Typically at  $z \gtrsim 4$ , the IGM becomes increasingly neutral and opaque to  $\text{Ly}\alpha$  photons, leading to attenuation of the line pro-

file. However, IGM cannot significantly attenuate the red peak without almost completely removing the blue peak. In our case, the persistence of a blue peak indicates that the IGM cannot be the primary cause of the red peak narrowing at high redshift. Instead, the narrowing is more likely driven by intrinsic galaxy/CGM evolution (e.g., lower outflow velocity dispersion, lower H I column density in outflows, or more ionised outflows at high redshift).

We also note that the spectral resolution of MUSE improves with wavelength, from a line spread function (LSF) of approximately  $170 \text{ km s}^{-1}$  at  $4700 \text{ \AA}$  to about  $90 \text{ km s}^{-1}$  at  $9350 \text{ \AA}$ . This enhances resolution at longer wavelengths and makes it easier to resolve and detect narrow features, particularly the blue peak, at higher redshifts. However, despite this advantage, the consistent suppression of the blue peak's line width across redshift further underscores the strong and persistent impact of IGM attenuation on the blue side of  $\text{Ly}\alpha$  emission.

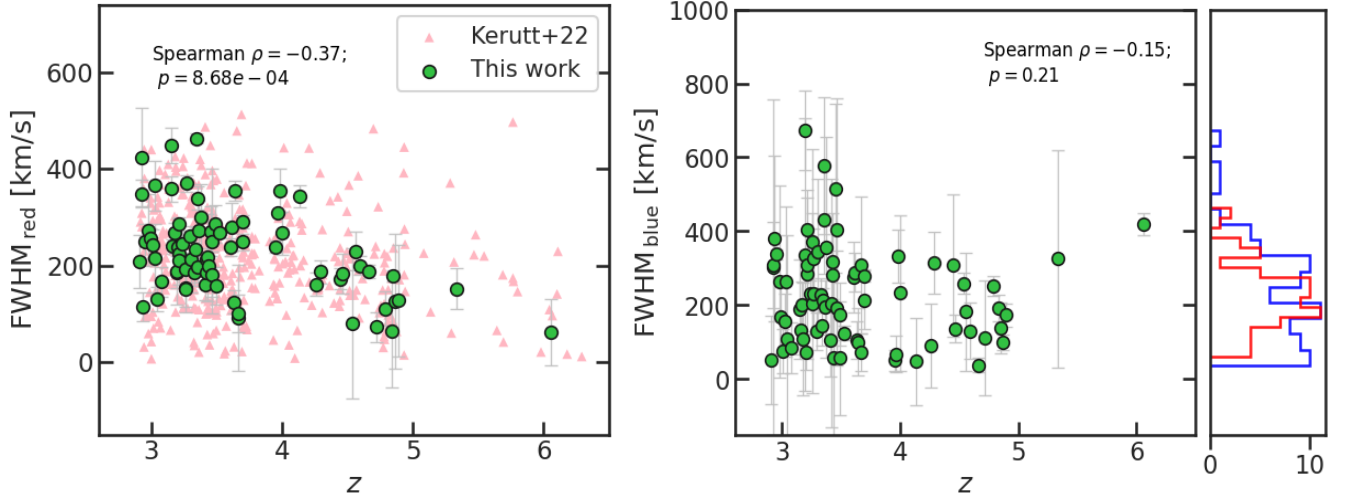
### 5.2.4 Correlations among velocities, peak separation and line width

Our conservative sample spans  $\text{Ly}\alpha$  peak separations of  $\Delta_{\text{peak}} = 223 - 1008 \text{ km s}^{-1}$ , with a mean of  $498.98 \pm 73.12 \text{ km s}^{-1}$ . This is consistent with previous findings from MUSE-Wide and MUSE-Deep LAEs ( $481 \pm 244 \text{ km s}^{-1}$ ; Kerutt et al., 2022), the MUSE Extremely Deep Field ( $534 \pm 28 \text{ km s}^{-1}$ ; Vitte et al., 2025), and  $z \sim 2.2$  LAEs ( $500 \pm 56 \text{ km s}^{-1}$ ; Hashimoto et al., 2015). Our sample remains above the minimum measurable threshold of  $150 \text{ km s}^{-1}$  reported by Vitte et al. (2025).

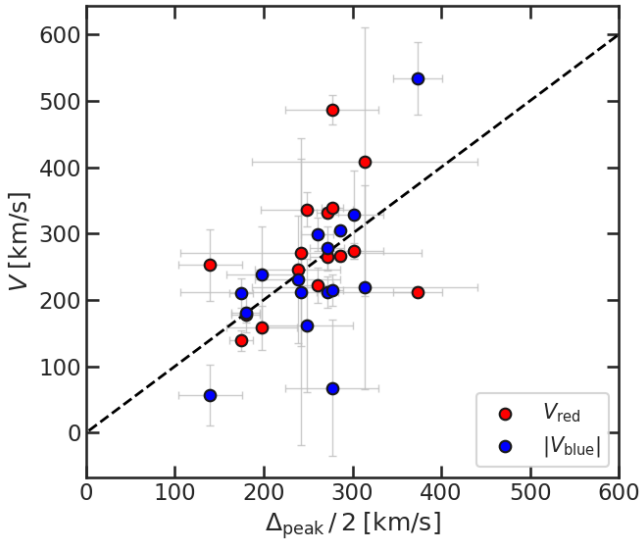
For LAEs with known systemic redshifts, we plot both the blue and red peak velocities against half of the peak separation ( $\Delta_{\text{peak}}/2$ ) in Figure 8, and we notice a strong correlation close to the one-to-one relation. This indicates that double-peaked profiles are typically symmetrically centered around the systemic velocity. Pahl et al. (2024) report that the ionising photon escape fraction,  $f_{\text{esc}}^{\text{LyC}}$  correlates more strongly with both  $\Delta_{\text{peak}}$  and  $V_{\text{blue}}$  than with  $V_{\text{red}}$  at  $z \sim 0.3$ . This suggests that the visibility of the blue peak in LAEs is closely linked to the escape of ionising photons, potentially indicating that similar neutral-phase gas dynamics govern both the escape of the blue  $\text{Ly}\alpha$  peak and the ionising photons.

We find a strong correlation between  $\text{FWHM}_{\text{red}}$  and  $\Delta_{\text{peak}}$  (see, Figure 9), indicating that broader red components are associated with larger peak separations. This trend is consistent with both observations (Yang et al., 2017; Verhamme et al., 2018; Kerutt et al., 2022) and radiative transfer models (Verhamme et al., 2015). To carry out a linear regression, we merge our dataset with those from Kerutt et al. (2022), Yang et al. (2017), and Verhamme et al. (2018), and use LtsFit<sup>8</sup> (Cappellari et al., 2013), which accounts for uncertainties in both variables and incorporates intrinsic scatter in the data. The combined data show Spearman correlation coefficients of  $p\text{-value} = 6.4 \times 10^{-33}$ ,  $\rho = 0.48$ . The intrinsic scatter estimated from the fit is  $29.6 \text{ km s}^{-1}$ . The resulting best-fit, shown by

<sup>8</sup><https://pypi.org/project/ltsfit/>



**Figure 7.** Redshift distribution of the Ly $\alpha$  peak line widths in our conservative sample. The left panel shows the FWHM of the red peak ( $\text{FWHM}_{\text{red}}$ ), while the right panel presents the FWHM of the blue peak ( $\text{FWHM}_{\text{blue}}$ ), both obtained from DAG fitting. In the left panel, the pink triangle markers represent LAEs from the MUSE-Wide and MUSE-Deep surveys (Kerutt et al., 2022).

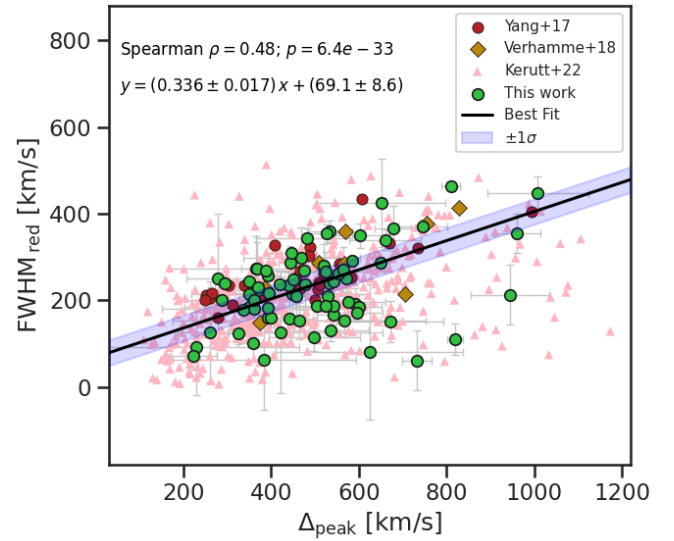


**Figure 8.** Velocities of Ly $\alpha$  blue and red peak are plotted against half of the peak separation ( $\Delta_{\text{peak}}/2$ ) for LAEs with known systemic redshifts in our sample. The black dashed line shows the one-to-one relation.

the solid black line in Figure 9, is given by

$$\text{FWHM}_{\text{red}} = (0.336 \pm 0.017)\Delta_{\text{peak}} + (69.1 \pm 8.6) \quad (4)$$

This result implies that redshifted Ly $\alpha$  emission broadens as it escapes further from the systemic velocity. This behavior has been interpreted via radiative transfer model using homogeneous shell models (Verhamme et al., 2018), simulations considering anisotropic or bipolar outflows and inhomogeneous gas distributions (Zheng & Wallace, 2014), and multiphase clumpy media models (Gronke & Dijkstra, 2016). Together, these models predict a robust  $V_{\text{red}}-\text{FWHM}_{\text{red}}$  correlation independent of outflow geometry or kinematics.

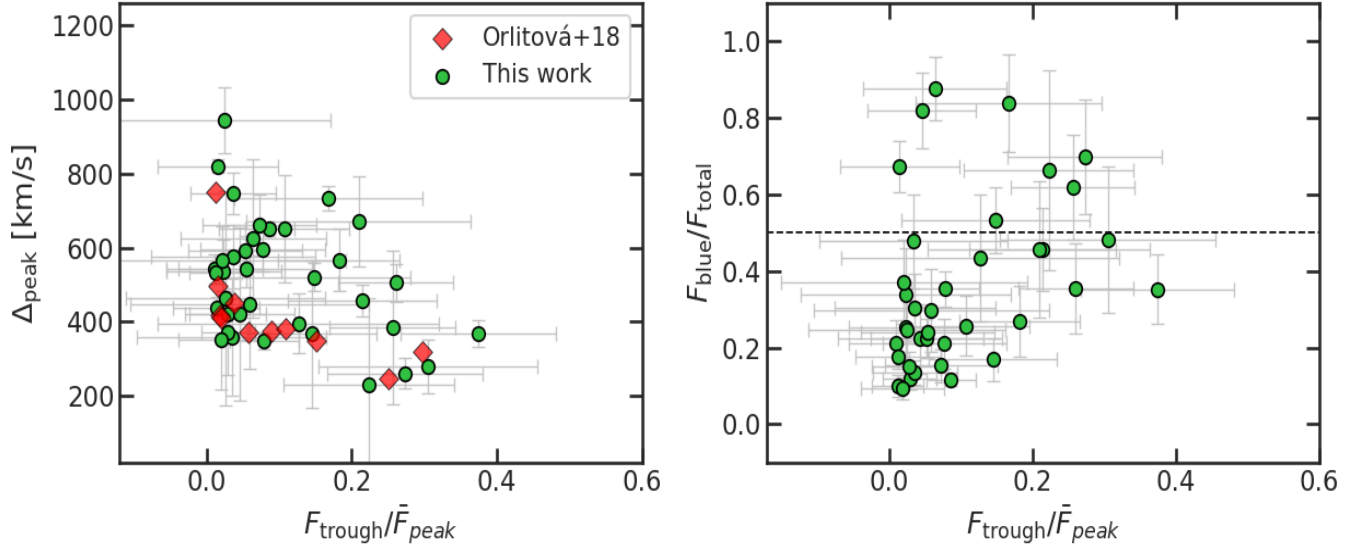


**Figure 9.** The FWHM of the main (red) peak is plotted against the double-peak separation of Ly $\alpha$ . Red circles represent  $z \sim 0.3$  Green Pea galaxies compiled by Yang et al. (2017). Yellow diamonds denote  $z = 3 - 4$  LAEs from Verhamme et al. (2018), using objects with peak separations listed in their Table 1., and pink triangles correspond to LAEs at  $z \sim 2.9 - 6.6$  from the MUSE-Wide and MUSE-Deep surveys (Kerutt et al., 2022). The best-fit to the combined data is shown as a black solid line, with the corresponding  $1\sigma$  confidence bounds indicated by blue dashed lines. The Spearman correlation coefficients and the equation of the best-fit line are displayed in the top left corner.

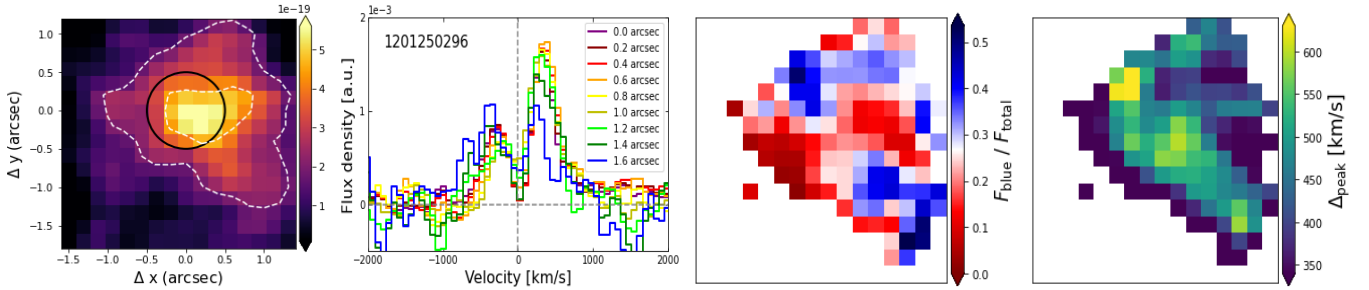
### 5.2.5 Trough flux density distribution

We find that several LAEs in our sample show positive residual flux in the absorption trough,  $F_{\text{trough}}$ , between the two peaks. The ratio of  $F_{\text{trough}}$  and the mean flux of the peak maxima,  $\bar{F}_{\text{peak}}$  serves as a diagnostic to distinguish clumpy from homogeneous shell models, showing different distributions in each (Gronke & Dijkstra, 2016). Throughout the rest of this





**Figure 10.** Ly $\alpha$  double peak separation (left panel) and blue-to-total flux ratio (right panel) versus normalised trough flux density  $F_{\text{trough}}/\bar{F}_{\text{peak}}$  for the conservative sample. Red diamonds are data points from Orlitová et al. (2018). Only sources with  $F_{\text{trough}}/\bar{F}_{\text{peak}} > 0.009$ , corresponding to the lowest value reported in Orlitová et al. (2018), are plotted. In the right panel, the horizontal black dashed line indicates the boundary between blue- and red-dominated regions.



**Figure 11.** Spatially resolved properties of one extended Ly $\alpha$  halo (ID: 1201250296) in our sample. 1st panel: Ly $\alpha$  pseudo narrow-band image; Coordinate is centred on the brightest pixel. Shown contours in white represent 2 and 4  $\sigma$  significance levels. The black circle denotes the location of stellar continuum detected in the MUSE white light image. 2nd panel: Spectra from annular bins as described in §5.3. 3rd panel: A pixel-by-pixel map of the blue-to-total flux ratio ( $F_{\text{blue}}/F_{\text{total}}$ ); and 4th panel: Peak separation map. 3rd and 4th panels are showing azimuthal variations in flux ratio and peak separation across the halo.

paper, we refer to this ratio ( $F_{\text{trough}}/\bar{F}_{\text{peak}}$ ) as the normalised trough flux density. Our result shows an anticorrelation between this ratio and peak separation (Figure 10, left panel), consistent with Orlitová et al. (2018) for green pea galaxies at  $z \sim 0.2 - 0.3$ . Such a correlation is absent in the clumpy model (Gronke & Dijkstra, 2016). While the clumpy model predicts ratios up to  $\sim 8$  with a peak near 0.3, our sample spans only between 0–0.4. We also find large scatter in peak separation for  $F_{\text{trough}}/\bar{F}_{\text{peak}} \sim 0$ , and a decrease to  $\lesssim 500 \text{ km s}^{-1}$  for  $F_{\text{trough}}/\bar{F}_{\text{peak}} \gtrsim 0.2$ , supporting the predictions from the homogeneous shell model (Orlitová et al., 2018).

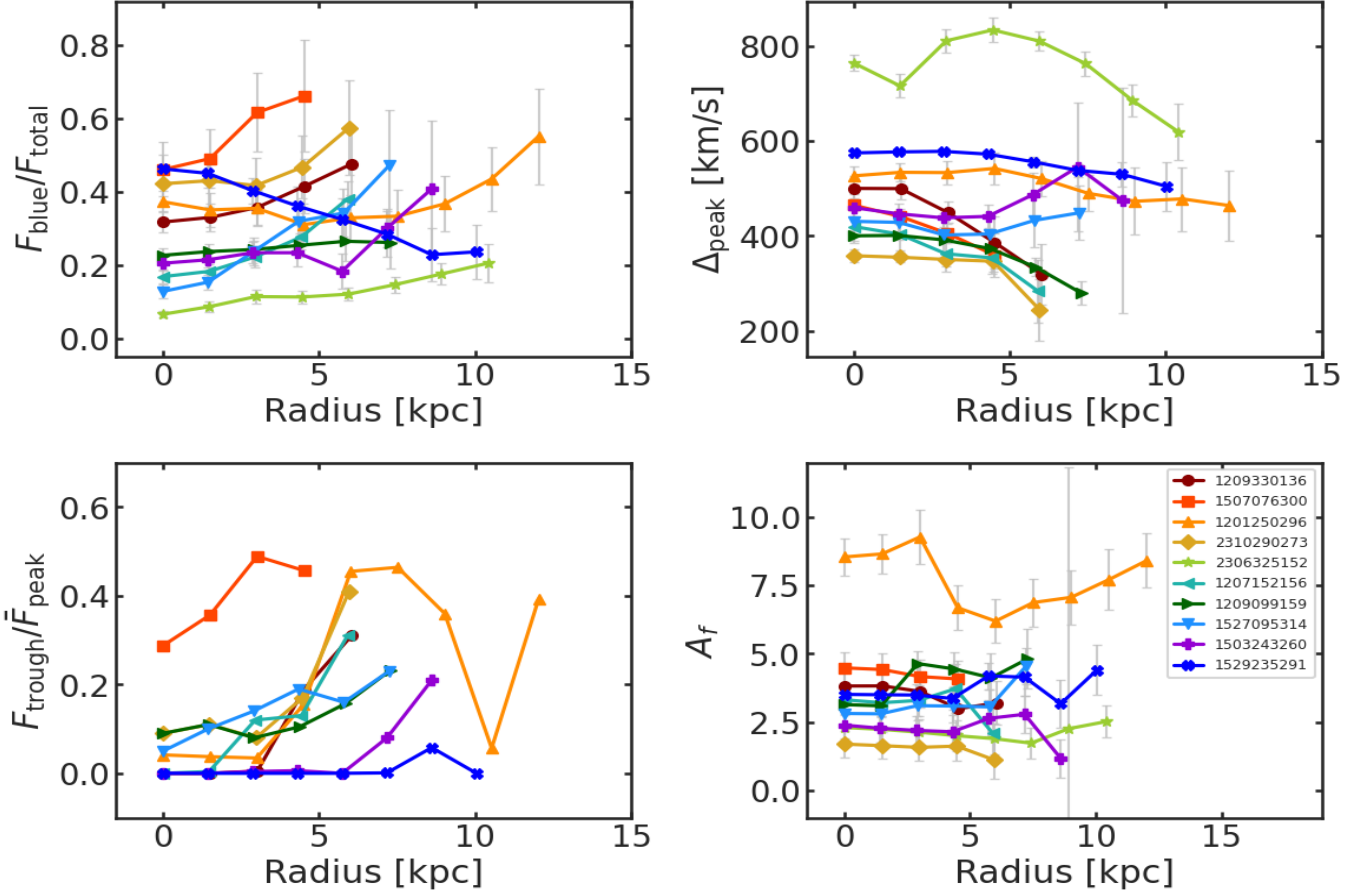
In Figure 10 (right panel), we plot  $F_{\text{blue}}/F_{\text{total}}$  against the  $F_{\text{trough}}/\bar{F}_{\text{peak}}$ , revealing distinct distributions in blue- and red-dominated regions. In red-dominated regions, trough flux rises with increasing blue peak flux. In contrast, strong blue-dominated LAEs ( $F_{\text{blue}}/F_{\text{total}} > 0.8$ ) show low trough flux ( $F_{\text{trough}}/\bar{F}_{\text{peak}} < 0.2$ ). This is likely due to the S/N of the

Ly $\alpha$  line, given that the blue-to-total flux ratio shows a strong dependence on  $(\text{S/N})_{\text{total}}$ , as discussed in §5.2.1.

### 5.3 Spatially resolved Ly $\alpha$ profiles

Integral field spectroscopy has enabled spatially resolved studies of LAEs. To investigate general trends and analyse the spectral properties of double-peaked LAEs as a function of radius, we employ a method similar to Erb et al. (2023). Among our 108 LAEs, we select 10 bright LAEs that show double-peaked profiles throughout their spatial extent with good S/N and exhibit clear spatial variations in their spectral profiles. They show extended Ly $\alpha$  halos in the continuum-subtracted narrowband images. While our selection ensures robust spatially resolved measurements, we note that double-peaked morphology does not necessarily persist at all radii in general. For example, Welton et al. (2024) report a halo with a single peak in the central region but double peaks in the inner and outer halo.

Annular spectra are created by binning spaxels radially.



**Figure 12.** Shown are results from line profile measurements of the spatially resolved spectra extracted from the annular regions described in §5.3. (top row) Left panel: blue-to-total flux ratio ( $F_{\text{blue}}/F_{\text{total}}$ ) as a function of radius (kpc), Right panel: Ly $\alpha$  peak separation versus radius; (bottom row) Left panel: normalised trough flux density ( $F_{\text{trough}}/\bar{F}_{\text{peak}}$ ) versus radius, Right panel: Ly $\alpha$  red peak asymmetry ( $A_f$ ) versus radius. MAGPI IDs corresponding to each galaxy are indicated in the Right panel of bottom row.

The high S/N of the Ly $\alpha$  halos enables reliable spatial binning. Starting from the central spaxel with the highest surface brightness, we include all spaxels with S/N > 2 in the continuum-subtracted Ly $\alpha$  narrowband images. Each halo is binned in radial increments corresponding to single spaxel (0.2''). The spectra within each radial bin are summed to construct Ly $\alpha$  profiles for different radii. We normalise these spectra by the total Ly $\alpha$  flux to clearly witness the variations in both peaks. We then fit a double-asymmetric Gaussian to these spectra via forward modeling, as in §4. In addition to analysing spatially averaged profiles, we also create pixel-by-pixel maps of the blue-to-total flux ratio and peak separation across the halos to investigate their azimuthal variations. Figure 11 shows an example of an extended Ly $\alpha$  halo in our sample, where the pseudo-NB image displays the extended emission. The spatially-resolved properties for all ten halos at  $z = 3.2 - 3.7$  are shown in Figure 18 in the Appendix.

Figure 12 presents the spatial variations of the spectroscopic quantities as a function of radius in kpc. Halo radii span 4.6–12 kpc. In 9 of 10 halos,  $F_{\text{blue}}/F_{\text{total}}$  increases with radius, consistent with  $z \sim 2$  double-peaked Ly $\alpha$  studies (Erb et al., 2023; Weldon et al., 2024), likely due to line-of-sight

effects in the outer halo showing more flux in the blue peak. In addition, peak separation decreases with radius, in agreement with the findings of Erb et al. (2023) and Weldon et al. (2024). However, we find that these spatial variations are not always spherically symmetric. Instead, significant azimuthal differences are present in some cases (see the example given in Figure 11, and also see Figure 18 in the Appendix), with certain regions within the halo exhibiting enhanced blue flux and reduced peak separation, rather than these features being uniformly distributed from the core to the outskirts. As a result, this azimuthal structure also somewhat influences the interpretations drawn from spatially averaged profiles based on annular regions. Azimuthal variations in the peak ratio suggest velocity asymmetries and nonradial gas motions at large radii. We also observe that the normalised trough flux density generally shows an increasing trend with radius, while Ly $\alpha$  red peak asymmetry shows a mild anticorrelation for some of them but no strong trend is observed. Generally, beyond  $\sim 4$  kpc, blue flux and the flux in the absorption trough rise while the peak separation falls from center to outward, though the small sample limits physical constraints on this scale.

We also see some exceptions in these trends. Interestingly,

one halo (ID: 1529235291) shows the opposite trend: a decreasing flux ratio with radius, while peak separation still decreases. Similar behavior is seen in blue-dominated halos (Mukherjee et al., 2023; Bolda et al., 2024), where the central regions with the highest surface brightness are blue-dominated and the outskirts show a stronger red peak. We also notice a moderate decrease in normalised trough flux density with radius for this source. Additionally, the halo with ID: 1503243260 shows a slight increase in peak separation with radius. We explore these trends in more detail in the next section in the context of spatially resolved radiative transfer modeling.

## 6. Discussions

In this section, we discuss the spectroscopic properties in the context of Ly $\alpha$  radiative transfer simulation at both global and local scales.

### 6.1 Radiative transfer modeling

Studies on Ly $\alpha$  radiative transfer through clumpy shells have shown that the Ly $\alpha$  spectral profiles can be more complex than those from homogeneous shells. In this work, we adopt the simple and widely used homogeneous shell model. We utilise the publicly available radiative transfer code zELDA<sup>9</sup> (Gurung-López et al., 2022) to model both spatially integrated and resolved double-peaked Ly $\alpha$  profiles. zELDA is based on radiative transfer code FLAREON (Gurung-López et al., 2019). This code employs a thin-shell geometry for the gas distribution surrounding the Ly $\alpha$  emitting region, assuming an isothermal, homogeneous, spherical shell of neutral hydrogen at  $10^4$  K, characterised by a uniform radial bulk velocity,  $V_{\text{shell}}$ . Importantly, zELDA accommodates both inflow ( $V_{\text{shell}} < 0$ ) and outflow ( $V_{\text{shell}} > 0$ ) scenarios, with emergent spectra from inflow is identical to that of a shell with an outflow velocity, but mirrored around the zero Ly $\alpha$  velocity (systemic). This framework enables an effective analysis of the gas dynamics within the ISM and CGM of the studied LAEs. The model incorporates five free parameters: shell velocity ( $V_{\text{shell}}$  in km s<sup>-1</sup>), neutral hydrogen column density ( $\log(N_{\text{HI}})$  in cm<sup>-2</sup>), dust optical depth ( $\tau$ ), intrinsic equivalent width ( $\log(EW)_{\text{int}}$  in Å), and intrinsic line width ( $W_{\text{int}}$  in Å). Ly $\alpha$  and continuum photons are generated from the central source with an intrinsic width of  $W_{\text{int}}$  and intrinsic equivalent width of  $\log(EW)_{\text{int}}$  and propagate through the H I shell, undergoing absorption or resonant scattering while travelling.

zELDA performs forward modeling that accounts for the instrument's resolution and pixelisation, and employs a Markov Chain Monte Carlo (MCMC) approach to fit the observed line profiles. The S/N of the spectrum is also incorporated. To estimate uncertainties, the shell parameters and redshift are derived from the observed spectrum after applying noise perturbations 1000 times. The best-fit models effectively reproduce key features of global double-peaked Ly $\alpha$  profiles. The best-fit H I column densities range from  $\log(N_{\text{HI}}) = 18.36 - 21.12$  cm<sup>-2</sup>,

which is consistent with the typical range observed in previous LAE studies.

### 6.2 Global spectroscopic properties and $N_{\text{HI}}$ dependence

Radiative transfer modeling of Ly $\alpha$  emission, using both simplified shell geometries and more complex clumpy configurations, has successfully reproduced observed correlations between spectral features and physical model parameters. For instance, a strong correlation has been found between the separation of the Ly $\alpha$  peaks and  $N_{\text{HI}}$  in the ISM and CGM of galaxies (see, Verhamme et al., 2015, 2018; Li & Gronke, 2022). A lower  $N_{\text{HI}}$  allows Ly $\alpha$  photons to escape with fewer scatterings, leading to a smaller peak separation. Additionally, the strength of the blue peak tends to decrease with increasing shell velocity (e.g., Verhamme et al., 2015; Li & Gronke, 2022; Erb et al., 2023).

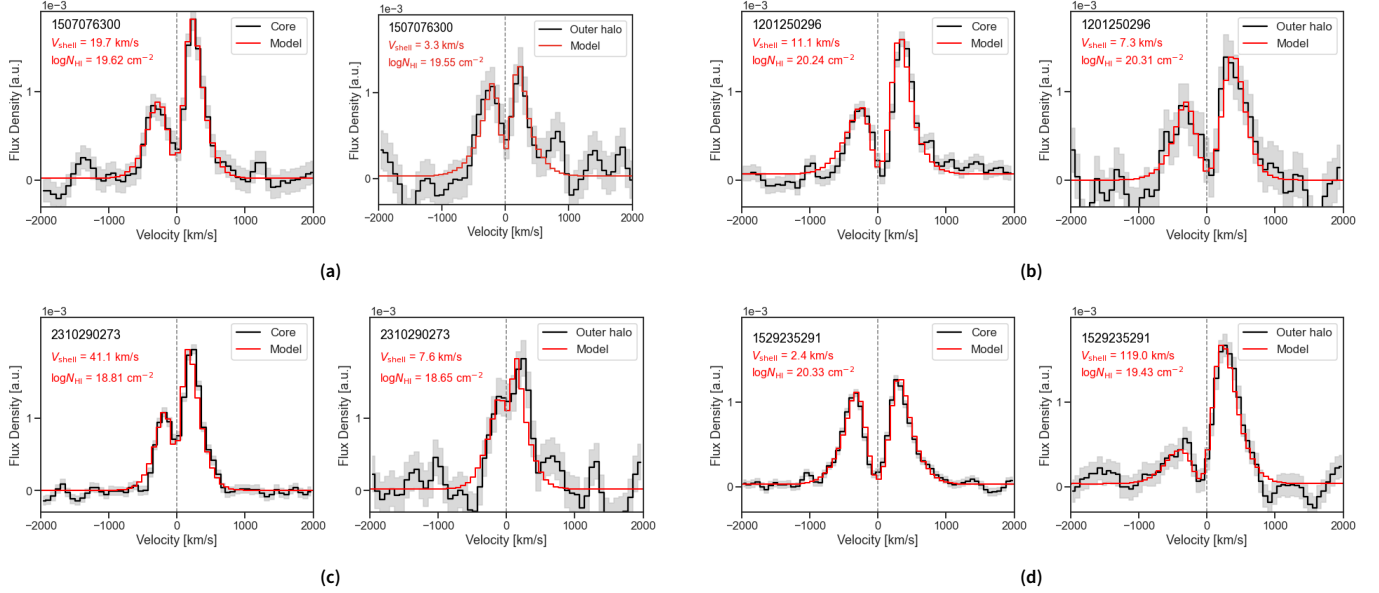
Furthermore, as discussed in § 5.2.4, the  $N_{\text{HI}}$  in the ISM and CGM primarily influences the escape of the red Ly $\alpha$  peak. A lower  $N_{\text{HI}}$  allows red Ly $\alpha$  photons to escape with less scattering, resulting in narrower lines and smaller peak separations. Conversely, at very high redshifts ( $z > 6$ ), LAEs within larger ionised bubbles exhibit broader red peaks (e.g., Mukherjee et al., 2024). The IGM predominantly absorbs the blue peak, diminishing its line flux, while the red peak is shifted out of the resonance frequency. Stacked Ly $\alpha$  spectra reveal a decreasing fraction of blue-side flux with increasing redshift (Hayes et al., 2021; Kramarenko et al., 2024), making double-peaked structures challenging to observe at high redshifts (Mason & Gronke, 2020). Consequently, the number of double-peaked LAEs decreases rapidly with redshift (see Figure 2), as the increasingly neutral IGM attenuates the blue peak, leading to single-peak detections. Ionised bubbles around high-redshift LAEs may allow blue peaks with small separations to shift out of resonance before encountering the IGM. Therefore, we observe minimal redshift effects on  $\text{FWHM}_{\text{blue}}$ . Kerutt et al. (2022) suggest that smaller peak separations correspond to smaller ionised bubbles necessary for the blue bump to shift sufficiently to survive IGM attenuation. More realistic radiative transfer models are required to further constrain these properties.

Given the anticorrelation between peak separation and  $F_{\text{trough}}/\bar{F}_{\text{peak}}$  in Figure 10, we expect that LAEs with relatively lower column densities typically show a higher trough flux. The higher the column density, the more scattering and absorption Ly $\alpha$  photons experience near the line center. This pushes Ly $\alpha$  photons further into the red and blue wings before they can escape. This results in greater suppression of the central flux, causing the trough between the peaks to deepen as  $N_{\text{HI}}$  increases. Therefore, a smaller peak separation and a higher trough flux strongly suggest the presence of low- $N_{\text{HI}}$  channels, which we will explore further in §6.4.

### 6.3 Spatially resolved modeling and variations across halos

To characterise the gas properties across the Ly $\alpha$  halos, we perform spatially resolved spectral modeling using zELDA. We fit spectra from the central 'core' region (within a  $0.2''$  radius

<sup>9</sup>[https://github.com/sidgurun/Lya\\_zELDA](https://github.com/sidgurun/Lya_zELDA)

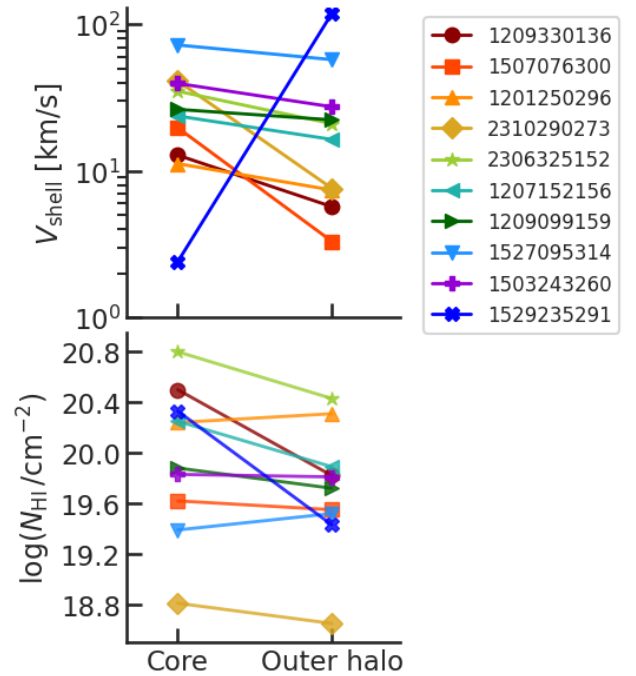


**Figure 13.** Examples of spatially resolved radiative transfer modeling of Ly $\alpha$  across four bright halos are shown. Observed spectra (black) and best-fit zELDA models (red) are presented for the ‘Core’ and ‘Outer halo’, with best-fit shell outflow velocities and H I column densities indicated in the top-left corner of each panel.

aperture) and the ‘outer halo’ region, selecting either the outermost annular bin or the preceding bin based on spectral quality. Modeling results for four bright halos are presented in Figure 13. zELDA successfully reproduces the observed double-peaked profiles in both the core and outer halo regions. The presence of a double-peaked profile in the cores indicates that Ly $\alpha$  photons experience multiple scatterings within the neutral gas shell. This contrasts with the halo studied by Weldon et al. (2024), where the central region appears single-peaked. They attribute this to backscattering, implying strong outflows that allow photons to escape without undergoing additional scatterings. In our case, the persistent double-peaked profile suggests that Ly $\alpha$  photons are more likely to undergo multiple scatterings before escaping.

For all ten halos, best-fit shell velocities and  $N_{\text{HI}}$  values for the core and outer halo regions are shown in Figure 14, revealing a clear decline in both parameters from the core to the outskirts. These findings align with previous studies (e.g., Erb et al., 2023; Weldon et al., 2024), supporting the notion that neutral gas velocity and  $N_{\text{HI}}$  regulate Ly $\alpha$  spectral variations across halos. The Ly $\alpha$  blue-to-total flux ratio is influenced by the line-of-sight velocity, which is largely a geometric effect arising from the decreasing projected component of the radial outflow velocity with increasing radius (Li et al., 2022). Weldon et al. (2024) argue that the decrease in the shell expansion velocity in the outermost regions of the Ly $\alpha$  halo may be driven by the increasing influence of gravitational deceleration at large radii. Additionally, the enhanced blue peak observed in the outer regions may also result from the preferential scattering of blueshifted Ly $\alpha$  photons to larger radii (Zheng et al., 2010). Peak separation trends are primarily driven by line-of-sight variations in  $N_{\text{HI}}$ , with lower column densities at larger radii allowing photons to escape closer to the line center without suffering much scattering, reducing

overall peak separation. Additionally, the observed trend of increasing residual flux in the absorption troughs with radius is mainly governed by the amount of residual neutral hydrogen in the ISM. A high  $N_{\text{HI}}$  indicates a larger amount of H I along the line of sight, which tend to absorb more Ly $\alpha$  photons near the line center, leading to a deeper trough between the peaks.



**Figure 14.** Results from the spatially resolved modeling showing the shell velocity ( $V_{\text{shell}}$ ) and H I column densities for two distinct regions: the core and the outer halo. MAGPI IDs are indicated in the legend, following the same order as in Figure 12.

Notably, MAGPI1529235291 shows an increase in shell



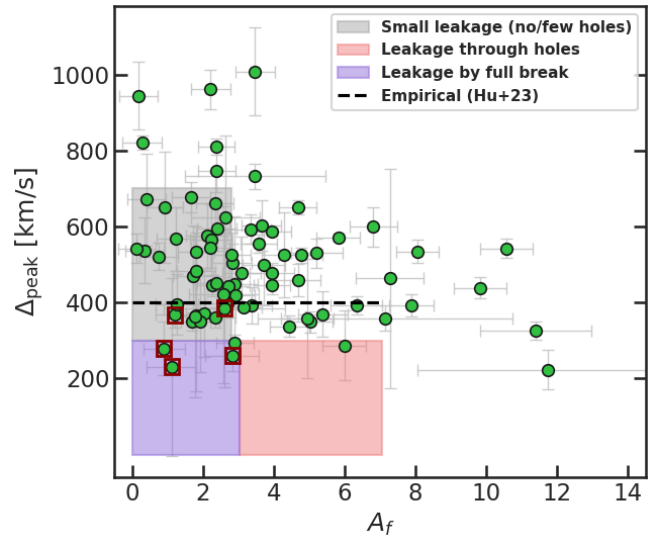
outflow velocity from the core to outer halo (panel d of Figure 13 and Figure 14), alongside a decreasing flux ratio with radius (Figure 12). Although its global and central Ly $\alpha$  spectra are red-dominated, the radial trend resembles that of a blue-dominated (accretion-dominated) system, suggesting active gas accretion in the central region and outflows in the outskirts. This LAE, with an integrated blue-to-total flux ratio of 0.42, exhibits a higher Ly $\alpha$  luminosity ( $6.92 \times 10^{42} \text{ erg s}^{-1}$ ) compared to others within the same ratio bin (see Figure 6), highlighting its distinct behaviour relative to typical outflow-dominated systems. In the future, spatially resolved studies of sources with comparable blue-to-total flux ratios could constrain the turnover point, beyond which the opposite trend is expected.

#### 6.4 Connection to global LyC escape

Previous studies suggest that significant Ly $\alpha$  flux near the systemic velocity may be a potential indicator of LyC escape (e.g., Rivera-Thorsen et al., 2017; Naidu et al., 2022). Given that many LAEs in our sample display positive residual flux at the absorption trough or close to the systemic velocity, we revisit the spatially integrated Ly $\alpha$  profiles to assess their potential as LyC leaker candidates. In the local universe, double peak separation correlates with LyC escape, with narrower peak separations generally associated with higher escape fractions (Verhamme et al., 2017; Izotov et al., 2018). Peak separation is a well-established parameter in the context of Ly $\alpha$  radiative transfer, and is directly linked to  $N_{\text{HI}}$  (Verhamme et al., 2015, 2017; Li & Gronke, 2022). However, peak separation alone does not uniquely reveal Ly $\alpha$  escape routes (Kakiichi & Gronke, 2021), as narrow separations can result from multiple small openings in a turbulent medium or a single large, wind-driven cavity (Hu et al., 2023). Instead, it can distinguish between different LyC leakage paths. Low LyC leakage, where only a few pathways allow LyC to escape, scattered Ly $\alpha$  photons traverse optically thick channels with high  $N_{\text{HI}}$ , resulting in a broad peak separation. In contrast, high leakage corresponds to narrow separations in density-bounded channels. However, at high redshifts ( $z > 2$ ), no clear correlation is observed between peak separation and LyC escape fraction (Kerutt et al., 2024).

We present the red peak asymmetry parameter,  $A_f$ , originally introduced by Rhoads et al. (2003). This parameter has recently been used to quantify the multiphase nature of turbulent H II regions and their connection to LyC escape (Kakiichi & Gronke, 2021; Hu et al., 2023).

Figure 15 illustrates the relationship between Ly $\alpha$  peak separation and red peak asymmetry, revealing an anticorrelation: profiles with strong asymmetry exhibit small peak separations, while a wider range of separations is observed for profiles with low asymmetry. When Ly $\alpha$  photons have direct escape routes through ionised, low-density channels, they experience fewer scatterings, leading to more symmetric profiles. Similarly, if the ISM is largely neutral but has a few clear, direct paths, the result is also relatively symmetric. However, in media with both density-bounded (low  $N_{\text{HI}}$ ) and optically thick neutral



**Figure 15.** Ly $\alpha$  double peak separation versus Ly $\alpha$  red peak asymmetry. The gray, blue, and pink shaded regions correspond to distinct LyC leakage regimes: minimal leakage (gray), escape through low  $N_{\text{HI}}$  channels (pink), and escape through large, density-bounded holes (blue). The black dashed line corresponds to the empirical boundary set by Hu et al. (2023). The brown squares mark 5 LAEs in our sample with  $F_{\text{trough}}/\bar{F}_{\text{peak}} \gtrsim 0.2$ , indicating they may be potential LyC leakers.

regions, some photons escape directly, while others scatter extensively within the dense, neutral gas, resulting in asymmetric profiles. In such a scenario, the scattered photons take longer paths, potentially emerging at different velocities than those escaping directly.

Following Hu et al. (2023), we classify the diagram (Figure 15) into three regions: (gray) minimal LyC leakage corresponding to systems with no or very few escape paths; (pink) significant leakage through low column density ionisation-bounded channels ( $f_{\text{esc}}^{\text{LyC}} > 10\%$ ); and (blue) substantial leakage through large holes density-bounded channels ( $f_{\text{esc}}^{\text{LyC}} > 10\%$ ). Based on this classification, we identify five LAEs as potential LyC leakers: four in the blue region ( $A_f < 3$ ), suggesting density-bounded channels escape, and one in the pink region ( $A_f > 3$ ), indicating ionisation-bounded leakage in a multiphase medium. Using the empirical relation between peak separation and  $f_{\text{esc}}^{\text{LyC}}$  from Izotov et al. (2018), these five LAEs have  $f_{\text{esc}}^{\text{LyC}} \sim 10 - 27\%$ . Many LAEs reside in the gray region, suggesting lower LyC escape fractions compared to those five LAEs in the blue and pink zones. Additionally, as discussed in Section 5.2.5, an anticorrelation exists between peak separation and  $F_{\text{trough}}/\bar{F}_{\text{peak}}$ . Three LAEs in the blue zone exhibit higher net Ly $\alpha$  flux in the trough (i.e.,  $F_{\text{trough}}/\bar{F}_{\text{peak}} > 0.2$ ), supporting the conclusion that these galaxies have LyC-thin sightlines (Verhamme et al., 2015; Gazagnes et al., 2020). Recent studies show that some high-redshift strong leakers exhibit large peak separations instead ( $\Delta_{\text{peak}} > 350 \text{ km s}^{-1}$ ; Fletcher et al., 2019; Naidu et al., 2022; Kerutt et al., 2024). Hu et al. (2023) has established an empirical boundary between the blue and

gray zones, located near a peak separation of  $400 \text{ km s}^{-1}$ . Applying this boundary, along with the trough flux criterion of  $F_{\text{trough}}/\bar{F}_{\text{peak}} > 0.2$ , we identify an additional two LAEs, resulting in a total of five LAEs as possible LyC leakers.

Cosmological hydrodynamic simulations highlight that the central escape fraction (fraction of the flux near the systemic velocity or absorption trough) is a more robust diagnostic of LyC leakage, whereas peak separation or asymmetry separately are insufficient indicators (Choustikov *et al.*, 2024). Given the diagnostic potential of double-peaked Ly $\alpha$  profiles in constraining CGM kinematics, column density, and LyC escape, expanding double-peaked sample will provide a more statistically significant dataset to further explore the physical mechanisms driving Ly $\alpha$  and LyC radiative transfer.

### 6.5 Caveats in modeling and analysis

Finally, we note several caveats, particularly concerning the modeling of Ly $\alpha$  profiles and the interpretation of the derived results. Our analysis employs a spherically symmetric radiative transfer model applied to annularly averaged spectra, assuming that all Ly $\alpha$  photons originate at the galaxy center and then scatter through an expanding or infalling shell before escaping into the CGM. This geometry is highly simplified, and the assumptions of central photon production and spherical symmetry are not fully consistent with the setup of annular averaging. In addition, the best-fit models are mainly driven by the high-S/N spectra from the innermost bins, while the outermost bins, though critical for tracing gradients in spectroscopic properties, provide weaker constraints on model parameters. Extracting physical insights from different halo regions will require more realistic anisotropic radiative transfer models with non-uniform H I distributions. Thus, the results presented here should be regarded as average parameters representative of the modeled regions rather than as precise local measurements.

## 7. Summary and conclusions

We present a comprehensive spectroscopic study of 108 double-peaked LAEs in the first 35 fields of the MAGPI survey, including spatially resolved analysis of ten extended Ly $\alpha$  halos. To interpret the observed features in the context of gas kinematics in galaxies, we conduct radiative transfer modeling on both global and spatially resolved spectra. Our main findings are summarised as follows:

- We apply an automated peak identification technique followed by spatial verification of double- and triple-peaked profiles, identifying 108 double-peaked LAEs. The number of double-peaked LAEs peaks at  $z = 3.2$  and drops sharply beyond  $z > 4$ , likely due to IGM attenuation. The double-peaked fraction at  $z < 4$  is  $\sim 37\%$ , consistent with previous such MUSE results at similar redshifts. Systemic redshifts are determined for 14 LAEs using C III] emission.
- We employ a forward modeling approach in which the intrinsic double-asymmetric Gaussian profile is convolved with the instrumental LSF to account for instrumental

broadening. This allows us to recover the intrinsic spectra prior to broadening.

- We find that 17% of the double-peaked LAEs are blue-dominated, indicative of gas accretion, while the rest are red-dominated, tracing outflows. The blue-to-total flux ratio strongly depends on the Ly $\alpha$  integrated total S/N. We observe an anti-correlation between Ly $\alpha$  luminosity and the blue-to-total flux ratio, with fainter LAEs showing a higher fraction of flux in the blue peak. However, the relation flattens at lower luminosities and higher flux ratios.
- The spectroscopic properties evolve with redshift: the red peak narrows significantly at  $z > 4$ , while the blue peak shows a weaker trend. Since the blue peak escape is strongly regulated by the IGM, its persistence implies that the red peak narrowing is instead driven by intrinsic galaxy/CGM evolution (e.g., reduced outflow velocity dispersion, lower H I column densities, or more ionized outflows). We therefore conclude that the H I column density in the ISM and/or CGM is the dominant factor shaping the Ly $\alpha$  red peak, independent of outflow geometry or kinematics.
- Several LAEs in our sample show residual trough flux between the two Ly $\alpha$  peaks. The normalised trough flux density anticorrelates with the peak separation, both tracing H I column density. The trough flux also correlates with the blue-to-total flux ratio, a trend largely driven by the total line S/N.
- We analyse spatially averaged Ly $\alpha$  profiles across galactic radii in ten LAEs with extended Ly $\alpha$  halos. Most of them show increasing blue-to-total flux ratios, normalized trough flux densities, and decreasing peak separations with radius. However, these trends are not always spherically symmetric—some systems exhibit significant azimuthal variations, with localized blue flux enhancements and reduced peak separations, complicating interpretations based on radial averages. Spatially-resolved radiative transfer modeling shows a clear line-of-sight effect: a declining shell expansion velocity and H I column density from center to outskirts, explaining the observed trends in blue-to-total flux ratio, peak separation and trough flux densities. One outlier shows increasing shell velocity and decreasing flux ratio in the outer halo. Though red-dominated, its kinematics resemble blue-dominated systems, suggesting a complex structure with central accretion and peripheral outflows.
- We observe an inverse correlation between peak separation and red peak asymmetry. Highly asymmetric red peaks are found to exhibit smaller peak separations. Using peak separation, red peak asymmetry, and normalised trough flux, we identify 5 LAEs in our sample as strong LyC leaker candidates.

In general, we find that double-peaked Ly $\alpha$  profiles, both globally and locally, consistently trace the kinematics and distribution of neutral hydrogen. This demonstrates the power of such profiles in probing the gas dynamics and ionising photon escape in high-redshift galaxies. In the future, a more robust statistical analysis of the double-peak fraction and its

redshift evolution will help to further quantify these trends. Additionally, advanced radiative transfer modeling that relaxes the assumption of azimuthal symmetry could provide deeper insight into the complex interplay between double-peaked profiles, gas kinematics, and escape of ionising photons in realistic, anisotropic environments in extended Ly $\alpha$  halos.

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**Data Availability Statement:** All MUSE data utilised in this work are publicly accessible through the ESO archive (<http://archive.eso.org/cms.html>). Data products such as fully reduced data cubes and emission line fits will be made available as part of an upcoming MAGPI team data release (Mendel et al. in prep; Battisti et al. in prep).

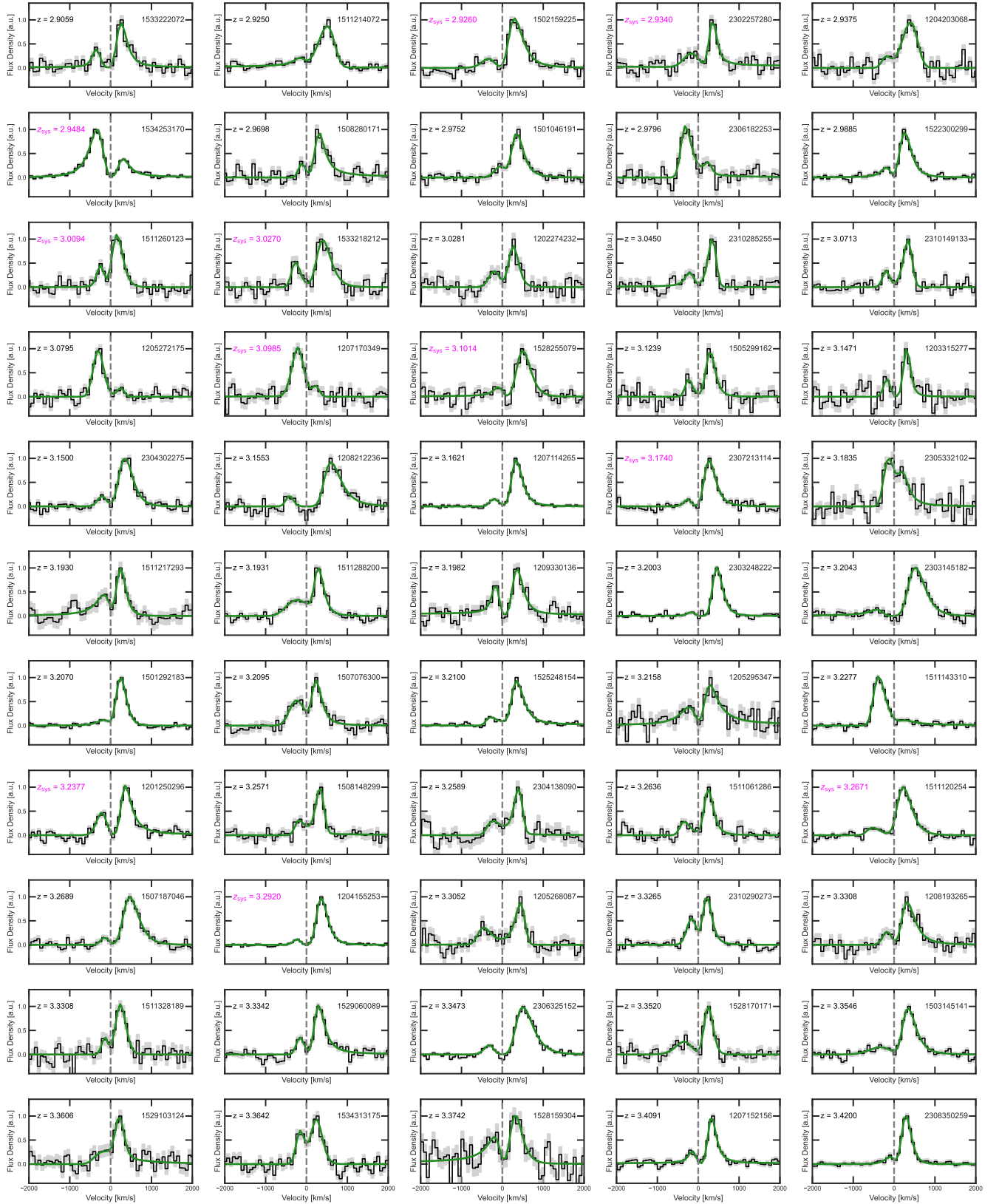
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## Appendix 1.





**Figure 16.** Double-asymmetric Gaussian (DAG) fitting to each line profiles as discussed in §4. The original spectra are plotted in black, while the best-fit convolved models are shown in green. The corresponding MAGPI IDs are shown in the top right corner, and the redshifts are displayed in the top left corner of each frame. For LAEs with a known systemic redshift, the redshifts are highlighted in magenta. The frames are arranged in ascending order of redshift. (Continued)

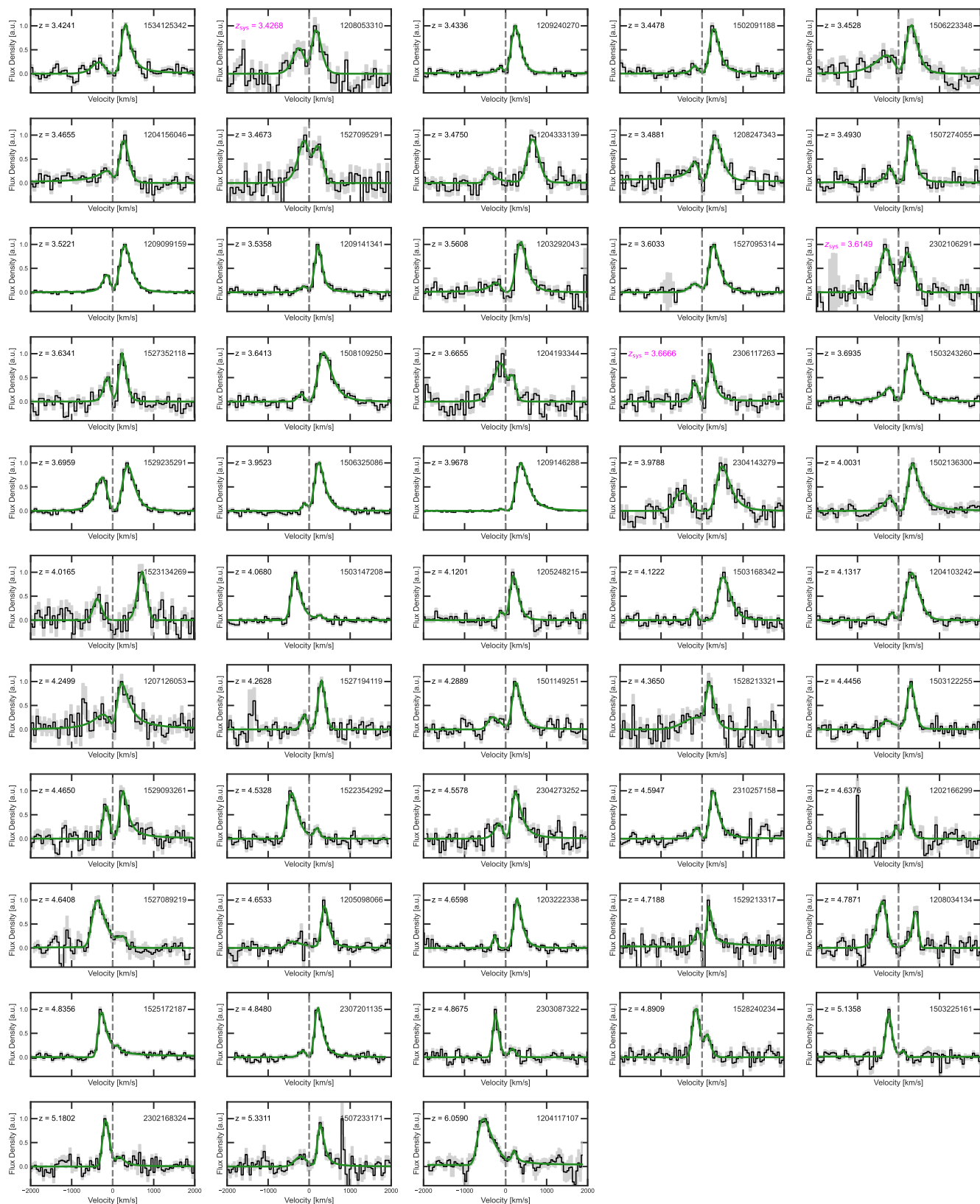
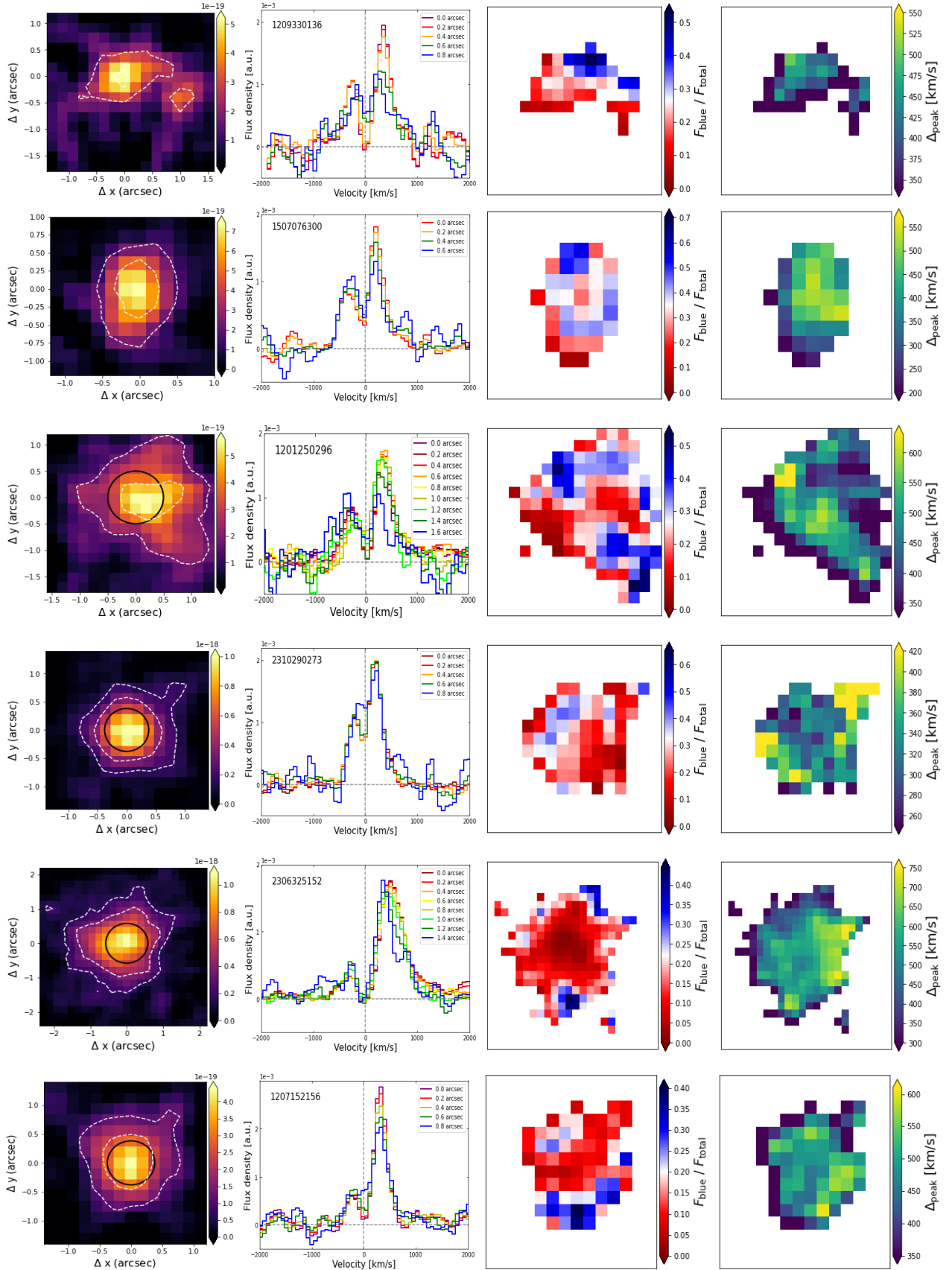


Figure 17. Same as Figure 16.



**Figure 18.** Spatially resolved properties of extended Ly $\alpha$  halos in our sample. 1st panel: Ly $\alpha$  pseudo narrow-band image; Coordinates are centred on the brightest pixel. Shown contours represent 2 and 4  $\sigma$  significance levels. The black circles denote the location of stellar continuum detected in the MUSE white light image. 2nd panel: Spectra from annular bins as described in §5.3. Corresponding MAGPI IDs are shown in the top left corner. 3rd panel: blue-to-total flux ratio ( $F_{\text{blue}}/F_{\text{total}}$ ) map. 4th panel: Peak separation map. (Continues...)

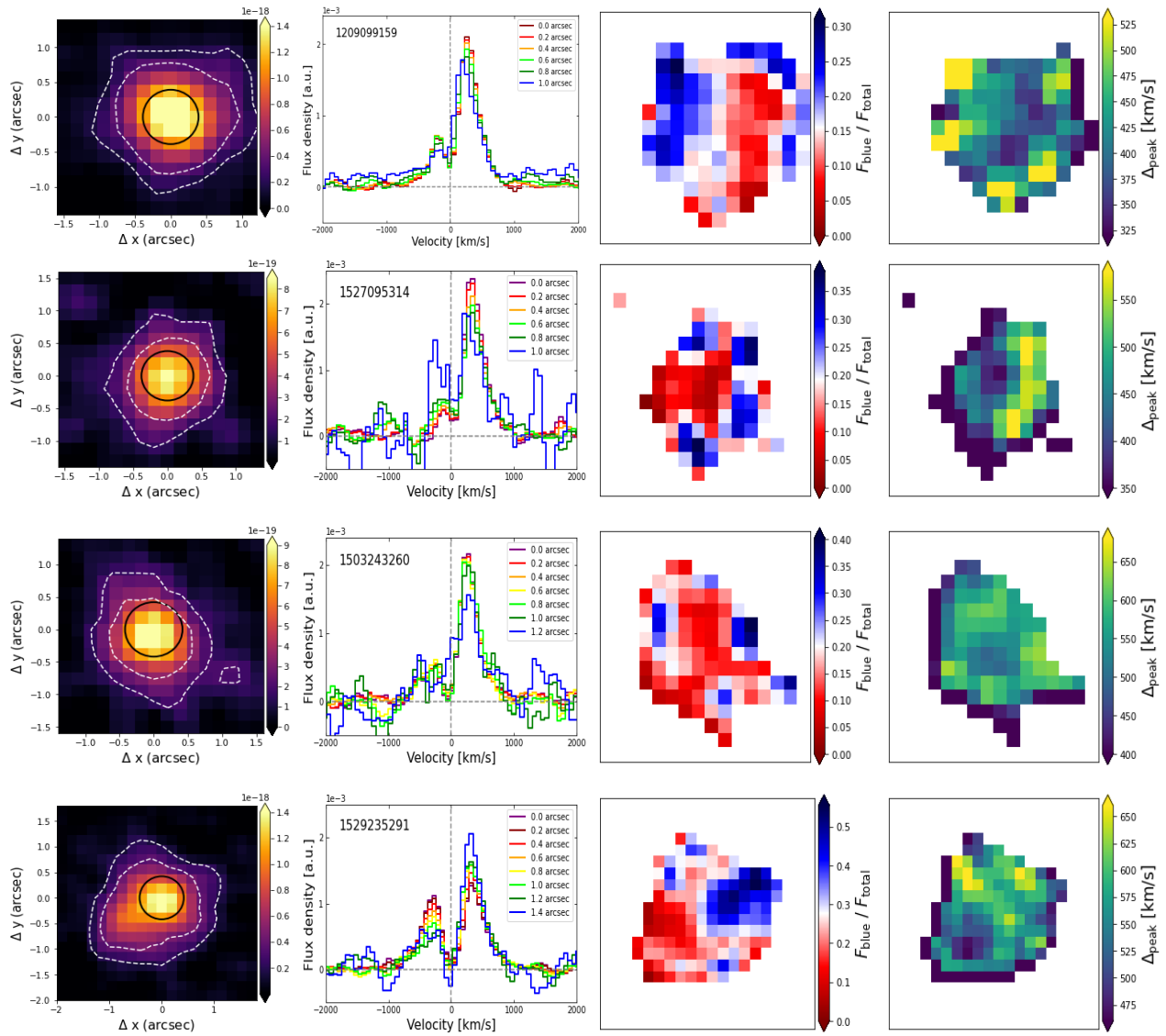


Figure 19. Same as Figure 18.



**Table 2.** Spectroscopic properties of 110 MAGPI double-peaked LAEs in the sample, in order of increasing redshift. Columns are as follows: Serial No.; MAGPI ID;  $z$ : Redshift;  $\log(L_{\text{Ly}\alpha})$ : Ly $\alpha$  luminosity in  $\text{erg s}^{-1}$ ;  $F_{\text{blue}}/F_{\text{total}}$ : Blue-to-total flux ratio;  $\Delta_{\text{peak}}$ : peak velocity separation, in  $\text{km s}^{-1}$ ;  $\text{FWHM}_{\text{blue}}$ : line width (FWHM) of blue Ly $\alpha$  peak, in  $\text{km s}^{-1}$ ;  $\text{FWHM}_{\text{red}}$ : FWHM of red Ly $\alpha$  peak, in  $\text{km s}^{-1}$ ;  $F_{\text{trough}}/\bar{F}_{\text{peak}}$ : normalised trough flux density;  $A_f$ : Ly $\alpha$  red peak asymmetry.

| Sl. No. | MAGPI ID   | $z$    | $\log(L_{\text{Ly}\alpha})$<br>[ $\text{erg s}^{-1}$ ] | $F_{\text{blue}}/F_{\text{total}}$ | $\Delta_{\text{peak}}$<br>[ $\text{km s}^{-1}$ ] | $\text{FWHM}_{\text{blue}}$<br>[ $\text{km s}^{-1}$ ] | $\text{FWHM}_{\text{red}}$<br>[ $\text{km s}^{-1}$ ] | $F_{\text{trough}}/\bar{F}_{\text{peak}}$ | $A_f$             |
|---------|------------|--------|--------------------------------------------------------|------------------------------------|--------------------------------------------------|-------------------------------------------------------|------------------------------------------------------|-------------------------------------------|-------------------|
| 1       | 1533222072 | 2.9059 | $42.03 \pm 0.05$                                       | $0.20 \pm 0.07$                    | $529.9 \pm 39.4$                                 | $53.0 \pm 119.5$                                      | $209.0 \pm 54.7$                                     | $< 0.009$                                 | $5.21 \pm 0.721$  |
| 2       | 1511214072 | 2.9250 | $42.40 \pm 0.04$                                       | $0.26 \pm 0.08$                    | $651.3 \pm 145.9$                                | $303.1 \pm 454.8$                                     | $424.3 \pm 102.1$                                    | $0.107 \pm 0.090$                         | $0.93 \pm 0.514$  |
| 3       | 1502159225 | 2.9260 | $42.54 \pm 0.03$                                       | $0.14 \pm 0.05$                    | $603.1 \pm 67.5$                                 | $309.1 \pm 116.4$                                     | $349.3 \pm 28.9$                                     | $< 0.009$                                 | $3.65 \pm 0.554$  |
| 4       | 2302257280 | 2.9340 | $41.86 \pm 0.05$                                       | $0.28 \pm 0.10$                    | $498.0 \pm 103.8$                                | $380.0 \pm 223.3$                                     | $114.2 \pm 30.0$                                     | $< 0.009$                                 | $3.72 \pm 0.590$  |
| 5       | 1204203068 | 2.9375 | $42.12 \pm 0.06$                                       | $0.14 \pm 0.09$                    | $627.4 \pm 253.9$                                | $139.8 \pm 308.8$                                     | $392.1 \pm 119.1$                                    | $0.194 \pm 0.157$                         | $0.73 \pm 0.527$  |
| 6       | 1534253170 | 2.9484 | $42.65 \pm 0.01$                                       | $0.77 \pm 0.03$                    | $571.6 \pm 9.8$                                  | $337.6 \pm 14.2$                                      | $250.5 \pm 35.8$                                     | $< 0.009$                                 | $5.81 \pm 0.608$  |
| 7       | 1508280171 | 2.9698 | $42.03 \pm 0.06$                                       | $0.13 \pm 0.06$                    | $384.6 \pm 153.7$                                | $51.1 \pm 307.7$                                      | $215.5 \pm 66.1$                                     | $< 0.009$                                 | $5.59 \pm 0.652$  |
| 8       | 1501046191 | 2.9752 | $42.55 \pm 0.04$                                       | $0.17 \pm 0.06$                    | $366.8 \pm 199.4$                                | $263.5 \pm 260.3$                                     | $273.0 \pm 74.4$                                     | $0.145 \pm 0.089$                         | $1.81 \pm 0.532$  |
| 9       | 2306182253 | 2.9796 | $41.94 \pm 0.05$                                       | $0.83 \pm 0.12$                    | $573.9 \pm 179.7$                                | $262.2 \pm 64.6$                                      | $111.4 \pm 269.6$                                    | $0.165 \pm 0.133$                         | $1.31 \pm 0.737$  |
| 10      | 1522300299 | 2.9885 | $42.45 \pm 0.02$                                       | $0.16 \pm 0.05$                    | $393.0 \pm 28.9$                                 | $168.6 \pm 97.6$                                      | $257.4 \pm 20.6$                                     | $< 0.009$                                 | $7.88 \pm 0.633$  |
| 11      | 1511260123 | 3.0094 | $42.26 \pm 0.04$                                       | $0.23 \pm 0.06$                    | $349.2 \pm 27.4$                                 | $75.1 \pm 59.9$                                       | $243.3 \pm 30.8$                                     | $< 0.009$                                 | $5.02 \pm 0.650$  |
| 12      | 1533218212 | 3.0270 | $42.07 \pm 0.05$                                       | $0.22 \pm 0.09$                    | $677.8 \pm 40.7$                                 | $155.1 \pm 75.2$                                      | $366.0 \pm 52.2$                                     | $0.044 \pm 0.114$                         | $1.66 \pm 0.545$  |
| 13      | 1202274232 | 3.0281 | $41.90 \pm 0.07$                                       | $0.30 \pm 0.11$                    | $448.8 \pm 177.4$                                | $264.4 \pm 202.2$                                     | $215.1 \pm 70.6$                                     | $0.059 \pm 0.164$                         | $2.87 \pm 0.664$  |
| 14      | 2310285255 | 3.0450 | $42.00 \pm 0.05$                                       | $0.25 \pm 0.09$                    | $536.5 \pm 86.3$                                 | $109.7 \pm 279.3$                                     | $131.6 \pm 25.1$                                     | $0.023 \pm 0.079$                         | $0.36 \pm 0.542$  |
| 15      | 2310149133 | 3.0713 | $42.14 \pm 0.04$                                       | $0.21 \pm 0.06$                    | $542.7 \pm 38.3$                                 | $83.8 \pm 68.3$                                       | $166.4 \pm 31.8$                                     | $0.011 \pm 0.067$                         | $0.13 \pm 0.525$  |
| 16      | 1205272175 | 3.0795 | $42.18 \pm 0.04$                                       | $0.88 \pm 0.11$                    | $565.0 \pm 251.8$                                | $247.6 \pm 111.6$                                     | $77.6 \pm 348.6$                                     | $0.038 \pm 0.087$                         | $0.84 \pm 0.941$  |
| 17      | 1207170349 | 3.0985 | $42.07 \pm 0.05$                                       | $0.91 \pm 0.14$                    | $483.9 \pm 270.7$                                | $283.6 \pm 147.4$                                     | $95.6 \pm 345.1$                                     | $0.128 \pm 0.136$                         | $0.18 \pm 0.706$  |
| 18      | 1528255079 | 3.1014 | $42.11 \pm 0.04$                                       | $0.10 \pm 0.06$                    | $554.0 \pm 104.8$                                | $159.1 \pm 332.0$                                     | $355.8 \pm 34.2$                                     | $< 0.009$                                 | $1.29 \pm 0.534$  |
| 19      | 1505299162 | 3.1239 | $41.98 \pm 0.05$                                       | $0.20 \pm 0.08$                    | $529.2 \pm 118.6$                                | $116.2 \pm 188.0$                                     | $193.7 \pm 63.6$                                     | $0.026 \pm 0.118$                         | $1.57 \pm 0.570$  |
| 20      | 1203315277 | 3.1471 | $41.83 \pm 0.09$                                       | $0.22 \pm 0.12$                    | $461.2 \pm 235.1$                                | $74.0 \pm 161.0$                                      | $100.8 \pm 39.0$                                     | $< 0.009$                                 | $3.92 \pm 1.950$  |
| 21      | 2304302275 | 3.1500 | $42.34 \pm 0.03$                                       | $0.11 \pm 0.04$                    | $534.3 \pm 49.7$                                 | $189.9 \pm 134.8$                                     | $359.2 \pm 26.3$                                     | $< 0.009$                                 | $1.80 \pm 0.524$  |
| 22      | 1208212236 | 3.1553 | $42.32 \pm 0.03$                                       | $0.10 \pm 0.03$                    | $1008.5 \pm 115.2$                               | $133.0 \pm 65.0$                                      | $448.6 \pm 36.5$                                     | $< 0.009$                                 | $3.47 \pm 0.564$  |
| 23      | 1207114265 | 3.1621 | $42.49 \pm 0.02$                                       | $0.10 \pm 0.03$                    | $534.3 \pm 31.2$                                 | $199.8 \pm 62.9$                                      | $239.6 \pm 15.0$                                     | $0.012 \pm 0.036$                         | $8.05 \pm 0.597$  |
| 24      | 2307213114 | 3.1740 | $42.01 \pm 0.03$                                       | $0.12 \pm 0.04$                    | $476.5 \pm 96.0$                                 | $107.4 \pm 152.1$                                     | $267.9 \pm 21.0$                                     | $< 0.009$                                 | $3.09 \pm 0.551$  |
| 25      | 2305332102 | 3.1835 | $42.02 \pm 0.08$                                       | $0.64 \pm 0.17$                    | $330.2 \pm 299.0$                                | $296.8 \pm 199.6$                                     | $207.4 \pm 268.5$                                    | $0.725 \pm 0.262$                         | $2.57 \pm 0.631$  |
| 26      | 1511217293 | 3.1930 | $41.97 \pm 0.06$                                       | $0.36 \pm 0.12$                    | $358.9 \pm 157.2$                                | $336.6 \pm 369.1$                                     | $185.0 \pm 64.7$                                     | $0.034 \pm 0.131$                         | $4.94 \pm 0.740$  |
| 27      | 1511288200 | 3.1931 | $42.42 \pm 0.06$                                       | $0.48 \pm 0.12$                    | $505.2 \pm 49.9$                                 | $673.8 \pm 107.0$                                     | $187.4 \pm 33.5$                                     | $0.261 \pm 0.080$                         | $2.82 \pm 0.562$  |
| 28      | 1209330136 | 3.1982 | $42.20 \pm 0.04$                                       | $0.31 \pm 0.07$                    | $477.5 \pm 26.8$                                 | $73.7 \pm 49.2$                                       | $238.1 \pm 59.2$                                     | $< 0.009$                                 | $3.93 \pm 0.625$  |
| 29      | 2303248222 | 3.2003 | $42.50 \pm 0.02$                                       | $0.04 \pm 0.02$                    | $585.5 \pm 305.0$                                | $52.5 \pm 351.0$                                      | $219.9 \pm 17.2$                                     | $< 0.009$                                 | $5.20 \pm 0.592$  |
| 30      | 2303145182 | 3.2043 | $42.39 \pm 0.02$                                       | $0.07 \pm 0.03$                    | $891.4 \pm 128.7$                                | $371.6 \pm 331.9$                                     | $451.5 \pm 26.5$                                     | $< 0.009$                                 | $2.10 \pm 0.528$  |
| 31      | 1501292183 | 3.2070 | $42.62 \pm 0.02$                                       | $0.12 \pm 0.03$                    | $370.5 \pm 114.5$                                | $284.1 \pm 228.7$                                     | $224.1 \pm 17.3$                                     | $0.030 \pm 0.037$                         | $2.04 \pm 0.518$  |
| 32      | 1507076300 | 3.2095 | $42.32 \pm 0.06$                                       | $0.46 \pm 0.11$                    | $458.4 \pm 42.7$                                 | $403.7 \pm 86.0$                                      | $211.1 \pm 50.4$                                     | $0.214 \pm 0.103$                         | $4.69 \pm 0.620$  |
| 33      | 1525248154 | 3.2100 | $42.73 \pm 0.01$                                       | $0.12 \pm 0.02$                    | $650.5 \pm 19.0$                                 | $307.7 \pm 94.4$                                      | $286.9 \pm 14.5$                                     | $0.087 \pm 0.034$                         | $4.68 \pm 0.525$  |
| 34      | 1205295347 | 3.2158 | $41.98 \pm 0.07$                                       | $0.26 \pm 0.11$                    | $455.9 \pm 182.8$                                | $399.2 \pm 369.7$                                     | $349.6 \pm 101.6$                                    | $0.019 \pm 0.196$                         | $5.58 \pm 0.694$  |
| 35      | 1511143310 | 3.2277 | $42.47 \pm 0.03$                                       | $0.88 \pm 0.07$                    | $674.3 \pm 531.4$                                | $289.4 \pm 41.8$                                      | $524.8 \pm 471.3$                                    | $0.034 \pm 0.066$                         | $4.09 \pm 0.671$  |
| 36      | 1201250296 | 3.2377 | $42.40 \pm 0.02$                                       | $0.28 \pm 0.03$                    | $542.7 \pm 24.4$                                 | $229.7 \pm 36.4$                                      | $244.7 \pm 27.3$                                     | $< 0.009$                                 | $10.56 \pm 0.730$ |
| 37      | 1508148299 | 3.2571 | $42.09 \pm 0.07$                                       | $0.27 \pm 0.09$                    | $566.9 \pm 83.9$                                 | $205.2 \pm 242.5$                                     | $154.7 \pm 40.3$                                     | $0.182 \pm 0.084$                         | $1.23 \pm 0.550$  |
| 38      | 2304138090 | 3.2589 | $41.98 \pm 0.09$                                       | $0.46 \pm 0.18$                    | $671.3 \pm 121.1$                                | $371.6 \pm 249.6$                                     | $151.0 \pm 47.6$                                     | $0.209 \pm 0.155$                         | $0.40 \pm 0.550$  |
| 39      | 1511061286 | 3.2636 | $42.27 \pm 0.03$                                       | $0.23 \pm 0.05$                    | $592.4 \pm 38.5$                                 | $229.8 \pm 101.6$                                     | $192.6 \pm 34.6$                                     | $0.053 \pm 0.078$                         | $3.34 \pm 0.587$  |
| 40      | 1511120254 | 3.2671 | $42.60 \pm 0.02$                                       | $0.13 \pm 0.03$                    | $746.3 \pm 55.4$                                 | $327.0 \pm 81.2$                                      | $370.2 \pm 16.8$                                     | $0.037 \pm 0.059$                         | $2.38 \pm 0.526$  |
| 41      | 1507187046 | 3.2689 | $42.36 \pm 0.02$                                       | $0.06 \pm 0.03$                    | $620.5 \pm 251.3$                                | $159.9 \pm 298.8$                                     | $407.8 \pm 36.5$                                     | $0.017 \pm 0.101$                         | $2.65 \pm 0.538$  |
| 42      | 1204155253 | 3.2920 | $42.68 \pm 0.01$                                       | $0.06 \pm 0.01$                    | $554.1 \pm 24.0$                                 | $129.5 \pm 50.7$                                      | $261.5 \pm 9.7$                                      | $< 0.009$                                 | $3.57 \pm 0.522$  |
| 43      | 1205268087 | 3.3052 | $42.02 \pm 0.08$                                       | $0.34 \pm 0.12$                    | $944.4 \pm 89.8$                                 | $343.4 \pm 198.1$                                     | $213.5 \pm 68.4$                                     | $0.024 \pm 0.146$                         | $0.18 \pm 0.531$  |
| 44      | 2310290273 | 3.3265 | $42.48 \pm 0.03$                                       | $0.35 \pm 0.05$                    | $349.4 \pm 23.2$                                 | $227.2 \pm 45.9$                                      | $185.2 \pm 27.1$                                     | $0.079 \pm 0.047$                         | $1.69 \pm 0.528$  |
| 45      | 1208193265 | 3.3308 | $42.14 \pm 0.05$                                       | $0.15 \pm 0.07$                    | $461.6 \pm 277.0$                                | $189.6 \pm 241.3$                                     | $334.6 \pm 88.3$                                     | $0.016 \pm 0.147$                         | $5.10 \pm 0.660$  |

Table 3. Same as in Table 2

| Sl. No. | MAGPI ID   | $z$    | $\log(L_{\text{Ly}\alpha})$<br>[erg s <sup>-1</sup> ] | $F_{\text{blue}}/F_{\text{total}}$ | $\Delta_{\text{peak}}$<br>[km s <sup>-1</sup> ] | $\text{FWHM}_{\text{blue}}$<br>[km s <sup>-1</sup> ] | $\text{FWHM}_{\text{red}}$<br>[km s <sup>-1</sup> ] | $F_{\text{trough}}/\bar{F}_{\text{peak}}$ | $A_f$         |
|---------|------------|--------|-------------------------------------------------------|------------------------------------|-------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|-------------------------------------------|---------------|
| 46      | 1511328189 | 3.3308 | 42.00 ± 0.05                                          | 0.16 ± 0.08                        | 376.3 ± 439.4                                   | 65.5 ± 316.6                                         | 246.2 ± 116.6                                       | < 0.009                                   | 2.77 ± 0.624  |
| 47      | 1529060089 | 3.3342 | 42.37 ± 0.02                                          | 0.18 ± 0.03                        | 438.6 ± 27.7                                    | 144.1 ± 54.5                                         | 233.9 ± 30.0                                        | 0.013 ± 0.060                             | 9.82 ± 0.737  |
| 48      | 2306325152 | 3.3473 | 42.79 ± 0.01                                          | 0.10 ± 0.01                        | 810.4 ± 21.1                                    | 214.5 ± 31.9                                         | 463.8 ± 10.9                                        | < 0.009                                   | 2.38 ± 0.513  |
| 49      | 1528170171 | 3.3520 | 42.09 ± 0.05                                          | 0.30 ± 0.09                        | 575.6 ± 85.1                                    | 431.3 ± 146.8                                        | 197.3 ± 31.1                                        | 0.036 ± 0.113                             | 2.11 ± 0.587  |
| 50      | 1503145141 | 3.3546 | 42.50 ± 0.03                                          | 0.15 ± 0.04                        | 660.1 ± 82.1                                    | 577.9 ± 186.1                                        | 340.2 ± 24.6                                        | 0.073 ± 0.079                             | 2.34 ± 0.528  |
| 51      | 1529103124 | 3.3606 | 41.95 ± 0.11                                          | 0.24 ± 0.16                        | 342.6 ± 485.9                                   | 821.3 ± 430.4                                        | 185.7 ± 98.9                                        | 0.383 ± 0.214                             | 2.78 ± 0.595  |
| 52      | 1534313175 | 3.3642 | 42.17 ± 0.05                                          | 0.35 ± 0.09                        | 368.0 ± 35.0                                    | 194.6 ± 73.1                                         | 272.5 ± 61.5                                        | 0.375 ± 0.107                             | 1.20 ± 0.543  |
| 53      | 1528159304 | 3.3742 | 41.93 ± 0.06                                          | 0.35 ± 0.12                        | 469.8 ± 132.0                                   | 356.1 ± 299.6                                        | 299.3 ± 69.8                                        | < 0.009                                   | 1.72 ± 0.571  |
| 54      | 1207152156 | 3.4091 | 42.19 ± 0.03                                          | 0.18 ± 0.04                        | 392.4 ± 47.6                                    | 106.3 ± 100.0                                        | 160.0 ± 31.0                                        | < 0.009                                   | 3.37 ± 0.560  |
| 55      | 2308350259 | 3.4200 | 42.71 ± 0.02                                          | 0.09 ± 0.03                        | 350.4 ± 132.6                                   | 205.7 ± 337.4                                        | 214.1 ± 23.1                                        | 0.019 ± 0.058                             | 1.92 ± 0.532  |
| 56      | 1534125342 | 3.4241 | 42.39 ± 0.03                                          | 0.19 ± 0.05                        | 599.6 ± 51.8                                    | 316.4 ± 100.6                                        | 186.3 ± 28.1                                        | < 0.009                                   | 6.81 ± 0.667  |
| 57      | 1208053310 | 3.4268 | 42.13 ± 0.09                                          | 0.43 ± 0.17                        | 396.0 ± 79.7                                    | 282.1 ± 166.5                                        | 217.3 ± 90.7                                        | 0.127 ± 0.194                             | 1.26 ± 0.609  |
| 58      | 1209240270 | 3.4336 | 42.43 ± 0.02                                          | 0.08 ± 0.03                        | 287.3 ± 92.4                                    | 57.6 ± 220.8                                         | 200.5 ± 19.2                                        | < 0.009                                   | 5.99 ± 0.607  |
| 59      | 1502091188 | 3.4478 | 42.36 ± 0.03                                          | 0.08 ± 0.04                        | 385.5 ± 338.8                                   | 59.4 ± 384.3                                         | 199.4 ± 40.8                                        | < 0.009                                   | 2.48 ± 0.557  |
| 60      | 1506223348 | 3.4528 | 42.27 ± 0.06                                          | 0.37 ± 0.11                        | 565.8 ± 92.8                                    | 516.0 ± 229.4                                        | 271.0 ± 52.0                                        | 0.021 ± 0.172                             | 2.24 ± 0.599  |
| 61      | 1204156046 | 3.4655 | 42.15 ± 0.05                                          | 0.22 ± 0.08                        | 362.1 ± 210.6                                   | 405.7 ± 355.2                                        | 181.7 ± 47.2                                        | < 0.009                                   | 1.77 ± 0.569  |
| 62      | 1527095291 | 3.4673 | 42.07 ± 0.10                                          | 0.48 ± 0.19                        | 279.2 ± 71.3                                    | 191.8 ± 156.1                                        | 250.9 ± 149.2                                       | 0.305 ± 0.151                             | 0.90 ± 0.585  |
| 63      | 1204333139 | 3.4750 | 42.09 ± 0.05                                          | 0.13 ± 0.09                        | 1088.3 ± 428.0                                  | 213.0 ± 338.0                                        | 287.4 ± 55.2                                        | 0.038 ± 0.160                             | 1.26 ± 0.546  |
| 64      | 1208247343 | 3.4881 | 42.32 ± 0.04                                          | 0.29 ± 0.08                        | 445.2 ± 27.6                                    | 175.8 ± 86.5                                         | 287.3 ± 39.0                                        | < 0.009                                   | 2.26 ± 0.548  |
| 65      | 1507274055 | 3.4930 | 42.16 ± 0.04                                          | 0.21 ± 0.06                        | 442.4 ± 53.2                                    | 56.5 ± 153.2                                         | 159.2 ± 39.0                                        | < 0.009                                   | 2.71 ± 0.601  |
| 66      | 1209099159 | 3.5221 | 42.82 ± 0.01                                          | 0.20 ± 0.01                        | 388.0 ± 7.9                                     | 125.0 ± 18.7                                         | 268.9 ± 9.2                                         | < 0.009                                   | 3.15 ± 0.520  |
| 67      | 1209141341 | 3.5358 | 42.46 ± 0.02                                          | 0.11 ± 0.04                        | 297.8 ± 250.7                                   | 132.7 ± 329.7                                        | 114.5 ± 23.6                                        | < 0.009                                   | 10.55 ± 0.880 |
| 68      | 1203292043 | 3.5608 | 42.54 ± 0.04                                          | 0.15 ± 0.07                        | 542.2 ± 125.0                                   | 258.4 ± 258.6                                        | 254.5 ± 27.6                                        | < 0.009                                   | 2.46 ± 0.607  |
| 69      | 1527095314 | 3.6033 | 42.58 ± 0.03                                          | 0.15 ± 0.04                        | 420.2 ± 39.6                                    | 274.5 ± 95.7                                         | 238.2 ± 17.9                                        | 0.029 ± 0.053                             | 2.93 ± 0.533  |
| 70      | 2302106291 | 3.6149 | 42.25 ± 0.04                                          | 0.53 ± 0.09                        | 521.0 ± 36.6                                    | 289.2 ± 51.4                                         | 279.2 ± 50.1                                        | 0.149 ± 0.131                             | 0.75 ± 0.546  |
| 71      | 1527352118 | 3.6341 | 42.47 ± 0.05                                          | 0.35 ± 0.07                        | 325.8 ± 24.2                                    | 106.8 ± 50.5                                         | 124.3 ± 24.2                                        | < 0.009                                   | 11.40 ± 1.573 |
| 72      | 1508109250 | 3.6413 | 42.46 ± 0.02                                          | 0.06 ± 0.02                        | 526.7 ± 58.3                                    | 100.8 ± 91.2                                         | 355.7 ± 20.6                                        | < 0.009                                   | 2.80 ± 0.537  |
| 73      | 1204193344 | 3.6655 | 42.23 ± 0.11                                          | 0.67 ± 0.26                        | 230.3 ± 234.8                                   | 309.0 ± 185.2                                        | 91.5 ± 110.6                                        | 0.223 ± 0.118                             | 1.11 ± 0.706  |
| 74      | 2306117263 | 3.6666 | 41.98 ± 0.09                                          | 0.25 ± 0.09                        | 359.3 ± 32.5                                    | 71.7 ± 50.9                                          | 101.3 ± 38.8                                        | < 0.009                                   | 7.14 ± 3.586  |
| 75      | 1503243260 | 3.6935 | 42.62 ± 0.02                                          | 0.18 ± 0.04                        | 450.0 ± 20.2                                    | 213.8 ± 78.8                                         | 248.9 ± 18.0                                        | < 0.009                                   | 2.37 ± 0.525  |
| 76      | 1529235291 | 3.6959 | 42.84 ± 0.01                                          | 0.42 ± 0.02                        | 586.1 ± 9.0                                     | 280.1 ± 17.6                                         | 290.5 ± 13.3                                        | < 0.009                                   | 3.95 ± 0.539  |
| 77      | 1506325086 | 3.9523 | 42.81 ± 0.02                                          | 0.10 ± 0.02                        | 293.4 ± 20.8                                    | 51.1 ± 27.6                                          | 238.5 ± 12.5                                        | < 0.009                                   | 2.90 ± 0.529  |
| 78      | 1209146288 | 3.9678 | 43.30 ± 0.00                                          | 0.02 ± 0.01                        | 446.0 ± 25.5                                    | 67.6 ± 49.8                                          | 308.8 ± 4.1                                         | < 0.009                                   | 3.94 ± 0.511  |
| 79      | 2304143279 | 3.9788 | 42.39 ± 0.04                                          | 0.32 ± 0.07                        | 961.1 ± 52.3                                    | 332.5 ± 72.4                                         | 355.9 ± 45.4                                        | < 0.009                                   | 2.22 ± 0.564  |
| 80      | 1502136300 | 4.0031 | 42.33 ± 0.03                                          | 0.13 ± 0.05                        | 525.4 ± 112.1                                   | 233.6 ± 210.5                                        | 267.2 ± 45.4                                        | < 0.009                                   | 4.27 ± 0.621  |
| 81      | 1523134269 | 4.0165 | 42.09 ± 0.11                                          | 0.35 ± 0.25                        | 1098.8 ± 54.2                                   | 193.0 ± 115.7                                        | 216.4 ± 45.7                                        | < 0.009                                   | 0.45 ± 0.552  |
| 82      | 1503147208 | 4.0680 | 42.59 ± 0.03                                          | 0.88 ± 0.08                        | 619.3 ± 483.3                                   | 205.8 ± 55.7                                         | 47.0 ± 215.4                                        | 0.060 ± 0.051                             | 3.23 ± 1.007  |
| 83      | 1205248215 | 4.1201 | 42.28 ± 0.04                                          | 0.10 ± 0.05                        | 277.2 ± 170.7                                   | 49.6 ± 153.7                                         | 182.4 ± 28.2                                        | < 0.009                                   | 1.48 ± 0.552  |
| 84      | 1503168342 | 4.1222 | 42.41 ± 0.03                                          | 0.08 ± 0.03                        | 690.0 ± 35.0                                    | 34.0 ± 41.3                                          | 314.9 ± 28.3                                        | < 0.009                                   | 2.09 ± 0.546  |
| 85      | 1204103242 | 4.1317 | 42.52 ± 0.02                                          | 0.06 ± 0.02                        | 482.3 ± 117.2                                   | 47.8 ± 116.9                                         | 343.7 ± 23.7                                        | < 0.009                                   | 1.80 ± 0.534  |
| 86      | 1207126053 | 4.2499 | 41.99 ± 0.06                                          | 0.20 ± 0.09                        | 417.2 ± 312.6                                   | 453.5 ± 376.8                                        | 306.7 ± 126.3                                       | 0.067 ± 0.236                             | 6.21 ± 0.718  |
| 87      | 1527194119 | 4.2628 | 42.24 ± 0.04                                          | 0.19 ± 0.07                        | 398.6 ± 39.0                                    | 89.8 ± 113.6                                         | 161.2 ± 24.4                                        | < 0.009                                   | 2.69 ± 0.640  |
| 88      | 1501149251 | 4.2889 | 42.36 ± 0.04                                          | 0.24 ± 0.07                        | 543.0 ± 50.8                                    | 314.4 ± 82.6                                         | 187.0 ± 23.8                                        | 0.054 ± 0.093                             | 2.20 ± 0.554  |
| 89      | 1528213321 | 4.3650 | 42.17 ± 0.09                                          | 0.32 ± 0.19                        | 306.1 ± 382.0                                   | 721.6 ± 274.7                                        | 181.8 ± 189.1                                       | 0.365 ± 0.254                             | 1.14 ± 0.581  |
| 90      | 1503122255 | 4.4456 | 42.40 ± 0.04                                          | 0.21 ± 0.07                        | 595.5 ± 59.6                                    | 307.6 ± 191.5                                        | 172.5 ± 22.3                                        | 0.077 ± 0.086                             | 2.39 ± 0.565  |

Table 4. Same as in Table 2

| Sl. No. | MAGPI ID   | $z$    | $\log(L_{\text{Ly}\alpha})$<br>[erg s <sup>-1</sup> ] | $F_{\text{blue}}/F_{\text{total}}$ | $\Delta_{\text{peak}}$<br>[km s <sup>-1</sup> ] | $\text{FWHM}_{\text{blue}}$<br>[km s <sup>-1</sup> ] | $\text{FWHM}_{\text{red}}$<br>[km s <sup>-1</sup> ] | $F_{\text{trough}}/\bar{F}_{\text{peak}}$ | $A_f$         |
|---------|------------|--------|-------------------------------------------------------|------------------------------------|-------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|-------------------------------------------|---------------|
| 91      | 1529093261 | 4.4650 | 42.26 ± 0.04                                          | 0.33 ± 0.07                        | 393.0 ± 24.7                                    | 137.0 ± 37.9                                         | 183.4 ± 41.9                                        | < 0.009                                   | 6.33 ± 0.734  |
| 92      | 1522354292 | 4.5328 | 42.79 ± 0.03                                          | 0.88 ± 0.08                        | 624.4 ± 214.1                                   | 258.1 ± 28.4                                         | 81.0 ± 155.3                                        | 0.064 ± 0.100                             | 2.64 ± 0.972  |
| 93      | 2304273252 | 4.5578 | 42.26 ± 0.04                                          | 0.20 ± 0.06                        | 369.6 ± 59.5                                    | 182.2 ± 158.9                                        | 229.6 ± 40.2                                        | < 0.009                                   | 5.38 ± 0.691  |
| 94      | 2310257158 | 4.5947 | 42.50 ± 0.02                                          | 0.13 ± 0.03                        | 360.6 ± 25.5                                    | 130.2 ± 76.2                                         | 200.3 ± 20.0                                        | < 0.009                                   | 2.35 ± 0.555  |
| 95      | 1202166299 | 4.6376 | 42.11 ± 0.04                                          | 0.16 ± 0.06                        | 241.6 ± 226.0                                   | 47.7 ± 197.8                                         | 103.0 ± 24.3                                        | < 0.009                                   | 5.90 ± 0.887  |
| 96      | 1527089219 | 4.6408 | 42.46 ± 0.05                                          | 0.75 ± 0.11                        | 650.6 ± 725.6                                   | 331.7 ± 45.6                                         | 227.4 ± 359.4                                       | 0.329 ± 0.153                             | 0.30 ± 0.634  |
| 97      | 1205098066 | 4.6533 | 42.62 ± 0.04                                          | 0.22 ± 0.08                        | 786.9 ± 107.6                                   | 411.7 ± 234.0                                        | 164.2 ± 28.5                                        | 0.046 ± 0.120                             | 1.82 ± 0.553  |
| 98      | 1203222338 | 4.6598 | 42.76 ± 0.02                                          | 0.10 ± 0.03                        | 525.0 ± 19.5                                    | 37.5 ± 19.1                                          | 187.8 ± 13.9                                        | < 0.009                                   | 4.76 ± 0.609  |
| 99      | 1529213317 | 4.7188 | 42.17 ± 0.04                                          | 0.26 ± 0.08                        | 223.5 ± 52.7                                    | 111.0 ± 154.5                                        | 72.4 ± 31.4                                         | < 0.009                                   | 11.72 ± 3.677 |
| 100     | 1208034134 | 4.7871 | 42.78 ± 0.03                                          | 0.67 ± 0.07                        | 820.2 ± 18.1                                    | 251.4 ± 27.2                                         | 110.4 ± 35.4                                        | 0.015 ± 0.083                             | 0.28 ± 0.562  |
| 101     | 1525172187 | 4.8356 | 42.56 ± 0.07                                          | 0.62 ± 0.14                        | 384.8 ± 208.5                                   | 191.6 ± 35.6                                         | 63.7 ± 116.8                                        | 0.256 ± 0.087                             | 2.60 ± 1.705  |
| 102     | 2307201135 | 4.8480 | 42.42 ± 0.02                                          | 0.09 ± 0.02                        | 337.3 ± 27.9                                    | 137.7 ± 69.0                                         | 179.0 ± 12.1                                        | < 0.009                                   | 4.44 ± 0.559  |
| 103     | 2303087322 | 4.8675 | 42.30 ± 0.04                                          | 0.82 ± 0.10                        | 420.9 ± 233.7                                   | 99.4 ± 20.6                                          | 126.2 ± 140.7                                       | 0.046 ± 0.075                             | 2.59 ± 0.981  |
| 104     | 1528240234 | 4.8909 | 42.30 ± 0.06                                          | 0.70 ± 0.15                        | 260.6 ± 41.8                                    | 174.0 ± 31.3                                         | 127.4 ± 116.7                                       | 0.273 ± 0.107                             | 2.82 ± 0.757  |
| 105     | 1503225161 | 5.1358 | 42.50 ± 0.04                                          | 0.90 ± 0.10                        | 364.1 ± 369.8                                   | 133.5 ± 23.0                                         | 122.3 ± 95.5                                        | 0.093 ± 0.077                             | 1.09 ± 1.433  |
| 106     | 2302168324 | 5.1802 | 42.36 ± 0.07                                          | 0.78 ± 0.17                        | 339.5 ± 523.0                                   | 156.3 ± 22.6                                         | 165.4 ± 151.5                                       | 0.126 ± 0.089                             | 8.18 ± 1.293  |
| 107     | 1507233171 | 5.3311 | 42.40 ± 0.05                                          | 0.25 ± 0.09                        | 463.4 ± 288.4                                   | 325.7 ± 293.4                                        | 152.5 ± 41.3                                        | 0.026 ± 0.137                             | 7.27 ± 0.955  |
| 108     | 1204117107 | 6.0590 | 42.88 ± 0.05                                          | 0.84 ± 0.13                        | 732.5 ± 32.7                                    | 418.7 ± 30.7                                         | 61.6 ± 69.4                                         | 0.168 ± 0.130                             | 3.47 ± 1.976  |